NIHR | National Institute for Health and Care Research

NIHR Doctoral Fellowship (DF)

Funding, Support or Development Opportunity	2025/303 NIHR Doctoral Award (Cohort1)

SUMMARY INFORMATION

Application Title	Characterising Gut Microbiome Alterations in ITP to Inform Future Microbiome-Targeted Treatments
Contracting Organisation	Imperial College London
Partner Organisation(s)	IMPERIAL COLLEGE HEALTHCARE NHS TRUST
Start Date	01/04/2026
End Date	31/03/2029
Duration (months)	36
% FTE Commitment	100%
Percentage of award time undertaking practice/clinical work	0%
Total Research Costs Requested	£409,063.63
Has this application been previously submitted to this or any other funding body?	No
Partnership Fellowship	Not Applicable

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LEAD APPLICANT & RESEARCH TEAM

Lead Applicant

Full Name	Ms Camelia Vladescu
Organisation	IMPERIAL COLLEGE HEALTHCARE NHS TRUST
Department	Haematology
Professional Background	Other Health or Care Practitioner
Detailed Professional Background	Clinical Research Practitioner
Professional Registration	Academy for Healthcare Science - Clinical Research Practitioner Registration number: 12878
ORCID	0000-0003-3324-6763
Research Degree Status	Not registered
Date of Registration	
Percentage FTE uptake of research degree	
Date of submission of research degree	
Date of award of research degree	

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Contextual Factors	Since joining the NHS in 2017, my focus has been on clinical research delivery with no protected academic time, so all research outputs were completed in my personal time. During my MRes (2019–2020), I worked full-time in an understaffed clinical trials team, balancing academic and clinical duties. This became harder after contracting COVID-19 in April 2020, leading to a three-week illness and long COVID symptoms requiring two years of follow-up. Despite maintaining full-time duties, my research capacity was limited. After my MRes and secondment as Trial Manager on a COVID-19 study, I was promoted to Team Leader of the Non-Malignant Haematology Research Unit. I expanded the team from two to seven, introduced standardized systems, and developed a training programme. After surgery in December 2021 and recovery, I focused on building the unit. By 2023, we achieved first-in-UK and first-in-Europe recruitment milestones and gained industry recognition. With the team running independently, I began developing my clinical academic career. In 2023, I was awarded the Imperial Health Charity/NIHR Imperial BRC Pre-Doctoral Fellowship, providing my first protected academic time. During this period, I set up the pilot study that informed this PhD project. To further develop my proposal, I applied for the NIHR Pre-Application Support Fund, which is now enabling me to focus on this application. I am one of the first research practitioners in the UK to gain national accreditation- an important step in professionalising the research delivery workforce. However, there are few practitioner role models in clinical academia, making the path less defined. While this brings an opportunity to help shape an emerging pathway, formal support structures remain limited. These contextual factors underscore the non-traditional nature of my career and the resilience, initiative, and leadership I have demonstrated in creating opportunities for research in parallel with service delivery.

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APPLICATION DETAILS

Scientific Abstract
Research Question: What is the contribution of gut microbiome dysbiosis in Immune Thrombocytopenia (ITP), and can targeting this dysbiosis provide a cost-effective therapeutic intervention that is acceptable to patients?

Background: ITP gained attention during the COVID-19 pandemic, when many cases were diagnosed following vaccination or infection, although it remains a significant autoimmune disorder in its own right. It is characterised by platelet destruction, causing spontaneous bleeding risks from minor nosebleeds to life-threatening brain haemorrhages. Patients also experience fatigue, cognitive impairment, and mental health challenges. Current treatments are costly, have significant side effects, and are often ineffective. Treatment decisions are hindered by limited understanding of ITP's underlying mechanisms.
The gut microbiome (the community of microorganisms in the digestive tract) helps regulate immune function and may influence autoimmune diseases, including ITP. Early studies suggest a link between gut microbiome alterations (dysbiosis) and ITP, but findings remain inconclusive.

Aims and objectives: This research aims to understand the role of the gut microbiome in ITP and use these insights to inform future therapeutic strategies.

Specific objectives include:
1. Characterizing gut microbiome differences between ITP patients and healthy controls.
2. Identifying microbiome profiles associated with disease parameters and treatment response.
3. Developing a translational framework for a feasibility trial of microbiome-targeted therapies in ITP.

Methods: This cross-sectional, case-control study will compare 30 adult ITP patients and 30 age- and sex-matched healthy controls from their households to minimise confounding factors such as diet and environment. The gut microbiome will be assessed through shotgun metagenomic sequencing and Nuclear Magnetic Resonance (NMR) spectroscopy. Clinical data, including platelet counts, bleeding severity, cognitive impairment, and fatigue, will be analysed for links to microbiome features. Patient input will guide the design of potential microbiome-based intervention.

Timelines for Delivery: This three-year study will begin with recruitment, sample collection, and a health economics review of ITP (Year 1). In Year 2 I will focus on microbiome data analysis. In Year 3 I will synthesise findings, consult patients, and co-design a feasibility study for microbiome-targeted therapy. Milestones include recruitment of 30 patient-control pairs in Year 1, data analysis by month 24, and a translational framework complete by project end.

Knowledge Mobilisation, Dissemination and Impact: I will share findings with help from the Patient Steering Committee through events, written materials, and UK Patient Support Organisation channels. I'll also share results with healthcare staff at local, UK, and global meetings. Expected impacts include better patient quality of life, reduced treatment burden, and lower NHS costs.

Research Inclusion: The study will include adults with ITP and related healthy controls. A demographically diverse Patient Steering Committee will continue to guide study delivery and dissemination. Recruitment will prioritise equity, diversity, and inclusion (EDI), ensuring representation across ethnicity, gender, and socioeconomic backgrounds. Translators will be engaged for participants who need them, and patient-facing materials will be translated.

Plain English Summary

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What is ITP and why am I doing this study?
Immune Thrombocytopenia (ITP) is a rare condition that can affect anyone. Doctors do not fully understand what causes it, but it sometimes starts after an infection or a vaccine, like COVID-19. In ITP the immune system (which normally fights off germs) mistakenly kills platelets. Platelets are tiny protective cells in the blood that help stop bleeding. Most people with ITP have it for the rest of their life. Without enough platelets, people can bleed easily, even without injury. Bleeding can be mild, like nosebleeds, or sometimes very serious (e.g. bleeding in the brain).Treatments exist, but they do not help everyone. They can cause side effects, cost the NHS a lot, and may need to be taken for years. ITP can also affect people's mental health, daily life, and ability to work.

Some people with ITP say they have gut problems and that their platelet numbers go up after taking antibiotics or eating fermented foods. These are known to change the gut microbiome (bugs that live in our gut and keep us healthy). Many have asked if ITP could be linked to gut health. I already have results from a small study I did, that show the gut microbiome of people with ITP may be different from healthy people. In this study I want to look more closely to understand the exact differences. I also want to see if changes in the gut are linked to:
• How bad ITP is (bleeding and platelet numbers)
• How people feel day to day (tiredness, brain fog, ability to work)
• How well ITP treatments are working

What Will I Do?
• Recruit 30 adults with ITP and 30 healthy family members
• Collect information about signs of illness, number of platelets and quality of life (how sick people are and how they feel in their daily lives)
• Ask for small samples of stool (poop), blood, and urine (wee). I will test these samples to find out what bugs live in the gut (by shotgun metagenomics) and what chemicals they make (by NMR). These are the are the best scientific methods to study the gut microbiome
• Look for differences between the gut microbiome test results of patients and of healthy people
• Look for links between test results and how people feel and respond to ITP treatment
• Share the results with people in the study and others with ITP. And ask for their ideas to help plan the next study. The next study, will test treatments to fix the gut microbiome based on what we find in this study.

Working with People with ITP
This study began because people with ITP kept asking about the gut. From the start, I worked with a diverse group of patients of different ages, ethnicities, and backgrounds. With support from the Imperial Patient Experience Research Centre, I set up a Patient Steering Committee to guide the study. They helped shape the questions and methods and will continue to support the project, interpret findings, and help share results in ways that are clear to patients.

Why This Research Matters?
If we find what negative role the gut microbiome plays in ITP, we can try to fix this. This could lead to:
• Less need for expensive long-term ITP medication
• People with ITP feeling better in everyday life
• NHS cost savings

How I will share what I learn
• Publish findings in science journals
• Talk about them at NHS and research meetings
• Organise an event to share results with people in the study
• Work with the UK ITP Patient Support Group (national patient charity) to share results in newsletters, webinars, and events

Working with People and Communities Summary

Meaningful involvement of people affected by immune thrombocytopenia (ITP) is central to this project's design, delivery, and impact. PPIE has been embedded throughout, building on strong foundations established during my Pre-Doctoral Fellowship.

This research was inspired by my conversations with over 400 patients in the ITP clinic at Hammersmith Hospital and at various patient events. Many reported gut issues and their lived experiences and questions sparked the central research question.

Before my pre-doctoral fellowship, I also consulted six patients (of different ages, ethnicities, genders, and socio-economic backgrounds) to understand feasibility and relevance of the study. In addition to endorsing the project, they advised reducing clinic visits by posting stool collection kits to patients' homes (recommendation now embedded in the protocol).

The UK ITP Support Association funded part of my pre-doctoral fellowship, highlighting the project's relevance and importance to patients.

During my pre-doctoral fellowship, I co-organised two ITP patient education events at Hammersmith Hospital, where I presented the microbiome research in plain language. Feedback showed strong interest: 18 of 26 attendees at the first event had considered a microbiome-ITP link, and all 13 patients at the second were open to trying a microbiome-based intervention.

To formalise patient input and embed inclusivity, I worked with the Patient Experience Research Centre (PERC) at Imperial College London and received tailored support from Ms Maria Piggin to establish a Patient Steering Committee. I created an application form capturing demographic details (e.g. ethnicity, socioeconomic background, education) to ensure a diverse group. The Committee now includes six patients (including Diane White, Deputy CEO of the ITP Support Association) and two family members. Maria attended our first meeting and provided guidance on inclusive facilitation, which I continue to apply. During my pre-doctoral year, we met to review the study, discuss findings, and co-develop next steps, including this PhD proposal. They contributed to inclusion criteria, recruitment strategy, patient engagement plans, and the plain English summary.

The Committee will remain central throughout the PhD. We will co-design a recruitment poster (pre-PhD) and a results summary poster (Year 3). Their input will inform data interpretation and dissemination. We will meet at least twice per year, and I will maintain regular individual contact via phone and email (see Gantt Chart).

Further details are provided in the Detailed Research Plan:
• Page 4 (Consultations with patients) outlines the results of consultations during a patient engagement event held during my pre-doctoral fellowship.
• Page 5 (Inclusion & Exclusion Criteria) shows how the Patient Steering Committee influenced inclusion criteria.
• Page 6 (Objective 3) describes how patients will be kept informed of study results and involved in shaping a future interventional study.
• Pages 8–9 (Working with people and communities & Inclusive research) provide detailed information on how I will work with people affected by ITP
• Page 9 (Knowledge mobilisation, dissemination and impact) explains how study findings will be disseminated, with input from patients at every stage.

I will evaluate PPIE impact using reflective logs and Committee feedback. Contributors will be reimbursed in line with NIHR guidance (see Budget).

SUPPORT AND MENTORSHIP

NIHR Doctoral Fellowship (DF)

Full Name	Professor Mary Wells
Organisation	IMPERIAL COLLEGE HEALTHCARE NHS TRUST
Email Address	mary.wells5@nhs.net
Role	Mentor
Description of Support Provided	Professor Mary Wells is the Deputy Director of Nursing (Research) at Imperial College Healthcare NHS Trust and a Professor of Practice in Cancer Nursing at Imperial College London. As a leading cancer care researcher, she has expertise in quantitative and qualitative research and has authored over 120 publications. She is an experienced PhD examiner, has supervised 5 PhD students to completion and currently acts as primary or secondary supervisor to 6 research fellows. In her NHS role, she leads the research strategy for nurses, midwives and Clinical Research Practitioners (CRPs) and has supported numerous NMAHPs to achieve fellowships at pre-doctoral, doctoral and post-doctoral levels. She is a member of the NIHR DCAF panel and reviews applications for several national and international research schemes. She is also the Impact and Metrics lead for the UK's Clinical Academic Roles Implementation Network (CARIN) and a member of the CNO's Research Transformation and Clinical Academic Nursing Professors Networks. I had the pleasure of meeting Professor Wells at a CATO conference in 2019, where her insights inspired me to consider a career as a clinical academic. I attended several meetings featuring her as a speaker, and her passion for independent research, particularly in supporting Nursing, Midwifery, and Allied Health Professionals (NMAHPs) as they pursue clinical academic careers, has been truly motivating. When I decided to apply for the pre-doctoral fellowship, I reached out to her for mentorship and guidance, and she was immediately encouraging and supportive. We have met regularly throughout the application processes for my pre-doctoral fellowship, the NIHR Pre-Application Support Fund, and now for this current opportunity. The knowledge, passion, and feedback she provides are invaluable. Professor Wells is incredibly approachable, making our communication seamless, and I feel both supported and inspired by her. Our excellent rapport has fostered a highly productive mentorship, which will be essential as I navigate my PhD and continue to build my clinical academic career. We plan to meet every 2-3 months to discuss my development, provide supportive challenges, and advise me on training, career opportunities, and progression. As I pursue the Doctoral Award, I am grateful for Professor Wells' continued guidance, which will be crucial in my transition to an independent clinical academic career.

Full Name	Mrs Demetra Georgiou
Organisation	IMPERIAL COLLEGE HEALTHCARE NHS TRUST
Email Address	demetra.georgiou@nhs.net

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Role	Mentor
Description of Support Provided	Demetra Georgiu is the Chair of the British Society of Genomic Medicine and Transformation Manager for North West London. I chose her as my mentor because of her extensive experience in healthcare innovation, service design, and her multidisciplinary approach to healthcare, which aligns with my aspirations to transform patient care through research, particularly in autoimmune conditions like ITP. Demetra's background as a Genetic Counsellor, her leadership in genomic medicine, and her work integrating digital solutions into healthcare offer invaluable insights for my career. As one of the first nationally accredited Clinical Research Practitioners (CRPs), I recognize that mentorship opportunities in my field are rare. The CRP role lacks a traditional academic career structure, and this presents a barrier for individuals like me who aspire to become clinical academics. Demetra's expertise in clinical genetics and service transformation, combined with her leadership in policy development and research collaborations, will provide critical guidance as I navigate these challenges. Her experience in service design, digital healthcare, and genomic medicine is particularly relevant as I explore integrating microbiome-based interventions into ITP treatment. Demetra's insights will be invaluable in ensuring that these interventions are not only scientifically rigorous but can also be practically and efficiently integrated into clinical practice. Additionally, her background in developing NICE guidelines and engaging with healthcare stakeholders will help me ensure that my research remains patient-centered and aligned with national healthcare priorities. Demetra's role as a transformation manager will also support the translational aspects of my work, particularly in designing and implementing intervention studies. Her guidance will be crucial in ensuring that new microbiome-based therapies are feasible and scalable within the NHS, which is essential for their integration into existing ITP treatment pathways. With Demetra's mentorship, I will strengthen both my research expertise and leadership skills, enabling me to promote more inclusive pathways for CRPs and others facing similar barriers to academic careers. Her support will be pivotal in helping me achieve my long-term goal of leading interventional trials that use microbiome-targeted therapies to improve patient outcomes in ITP. We will meet every two months throughout my PhD to review my progress, address challenges, and ensure that my research remains aligned with the needs of patients and clinicians. These regular check-ins will also facilitate my development as a clinical academic, helping me transition from a research delivery role to an independent researcher.

Full Name	Professor Nichola Cooper
Organisation	Imperial College London

NIHR Doctoral Fellowship (DF)

Email Address	n.cooper@imperial.ac.uk
Role	Primary Academic Supervisor
Description of Support Provided	Professor Nichola Cooper will continue in her role as my primary academic supervisor. We have worked together closely since 2017 when I joined her team as a junior Clinical Research Practitioner. It has been a privilege to grow within her group over the years- from coordinating her clinical trials to now leading the Non-Malignant Haematology Clinical Research Unit. Our long-standing collaboration and regular supervisory contact (including weekly clinical and research meetings) have created a highly productive and supportive environment, which will continue throughout my PhD. Prof Cooper is a Professor of Immune Haematology at Imperial College London and an academic consultant in the Centre for Haematology within the Department of Immunology and Inflammation. She co-leads the Immunology Theme of the NIHR Imperial Biomedical Research Centre (BRC). Her expertise in immune-mediated haematological disorders, particularly Immune Thrombocytopenia (ITP), aligns directly with the focus of my proposed research. She leads the largest clinical cohort of ITP patients in the UK and is the founder and co-chair of the European Research Consortium on ITP. This is particularly important for my PhD research as it provides me with access to an expansive patient cohort. Professor Cooper's leadership in the European Research Consortium on ITP ensures that I will be connected with a network of leading experts and collaborators in the field, enhancing the quality and scope of my work. Professor Cooper has authored numerous papers in prestigious international, peer-reviewed journals, including Nature Immunology, Blood, and the British Journal of Haematology, further solidifying her reputation as a leader in the field. She also serves as a trustee for the UK ITP Patient Support Association. Her active involvement in shaping ITP research policy, along with her international leadership in delivering clinical trials of immune modulators, including designing, developing, and interpreting industry trials, as well as creating an academic platform study for COVID-19, will provide valuable strategic insight as I develop my future interventional research programme. In addition to scientific and clinical guidance, Professor Cooper will provide support with data analysis interpretation, post-doctoral application, and career development. I will also benefit from her established collaborations with the Imperial Clinical Trials Unit and Imperial Business School, particularly for future studies requiring input on clinical trial design and health economics. Moreover, Professor Cooper's close ties with the pharmaceutical industry and her national and international collaborations will open up unique opportunities for networking and collaboration, which will be key to the success of my research. Professor Cooper has extensive supervisory experience. She has successfully supervised three PhD students to completion, as well as

	multiple MSc, BSc, postdoctoral, and clinical research fellows. Currently, she supervises two PhD students, two clinical research fellows, and two postdoctoral fellows. Her previous students have presented at national and international meetings and produced high-quality publications. Supervision will be delivered through: • Weekly individual meetings to review research progress, address challenges, and provide feedback on outputs. • Monthly joint supervisory meetings with my co-supervisor (Prof. Julian Marchesi). • Participation in weekly research group and pre-clinic meetings to stay informed about the latest research and clinical developments, and to receive feedback on progress, as well as interim and final data analysis. • Ongoing mentorship through her open-door policy, which has been a cornerstone of our productive working relationship since 2017. Our frequent interactions and open lines of communication have fostered a collaborative environment where ideas and feedback flow freely. This ongoing partnership has been instrumental in the success of our projects, as it allows us to address challenges swiftly and leverage each other's strengths. Professor Cooper's commitment to mentorship and her accessibility have made it easy to maintain a productive and positive working relationship. Given our track record of successful collaboration, I am certain that this strong foundation will support the continued success of my PhD studies and transition toward clinical academic independence.

Full Name	Professor Julian Marchesi
Organisation	Imperial College London
Email Address	j.marchesi@imperial.ac.uk
Role	Academic Supervisor
Description of Support Provided	Professor Marchesi, a leading authority in Digestive Health and the co-founder of the Microbiome Network, was the chair of the British Society of Gastroenterology's Gut Microbiota for Health Expert Panel from 2018-2021 and chaired the Microbiology Society's "Unlocking the Microbiome" working group and the subsequent publication. His research group is formed of 14 PhD students, 2 post-doctoral fellows and 2 research technicians, providing a highly supportive environment for training and collaboration. He has successfully supervised numerous BSc and MSc students since 2001. He has supervised 42 PhD students who have completed their theses in the 4-year time span. Over his 24-year career, he has authored over 280 journal articles and peer-reviewed publications and was a senior editor for the International Society for Microbial Ecology Journal. His expertise in microbiome-host interactions, particularly through the use of metagenomic and metabolomic approaches, is essential to my project, which applies these techniques to study immune thrombocytopenia (ITP). As a co-

	supervisor, Professor Marchesi will provide specialist guidance on the analysis and interpretation of metagenomics and metabolomics data, as well as input into the translational aspects of the research. I built a strong working relationship with Professor Marchesi and his lab during my pre-doctoral fellowship, regularly attending his group meetings and meeting with him monthly to review my progress. Since completing the pre-doctoral fellowship, we have remained in close contact, and our collaboration will continue seamlessly into the PhD. We will formally meet on a monthly basis, but I will continue to attend his lab meetings and benefit from his open-door policy—something that was invaluable throughout my pre-doctoral work and continues to provide me with regular access to advice and support. His extensive expertise in microbiome research, combined with his considerable experience in supervising PhD students and researchers, ensures a high level of academic guidance and support. Moreover, Prof Marchesi's strong connections with various microbiome groups and experts nationally and internationally will offer unique collaborative opportunities and insights. In addition to his methodological and analytical input, Professor Marchesi is actively supporting my career development. He has helped arrange a two-month placement with Professor Rob Knight's lab at UC San Diego, an internationally renowned centre for metagenomic research, which will further strengthen my analytical skills and global collaborations. Prof Marchesi's wide network of collaborators in microbiome research will be a valuable asset during the PhD and as I develop future postdoctoral plans. Prof Cooper and I started collaborating with Prof Marchesi before I began my pre-doctoral fellowship. Since starting the fellowship, we have been meeting monthly to discuss my progress and the research study. Like Prof Cooper, he maintains an open-door policy, which has made it very easy to communicate outside our scheduled discussions. This ongoing collaboration has fostered an excellent professional relationship among the three of us, built on open communication. In practice, supervision feels seamless; both professors are readily available to provide guidance while also encouraging my independence as a researcher. This balance fosters a supportive environment that I am confident will help me thrive in my PhD and beyond.

SUMMARY BUDGET

Justification of Costs
Lead and Co-applicant
My full-time salary over the 3-year PhD (Band 7 NHS) to enable me to manage and deliver all aspects of the project.

Other Staff Costs
Salary cost for a staff member (Band 5, 1.5 days/week for 1 year) in the Haematology CTU to support

recruitment, study visits, and trial documentation in line with GCP and NIHR standards. Given my extensive experience in research delivery since 2017, I do not require further development in operational tasks. Delegating these responsibilities will allow me to focus on acquiring new skills essential for the academic components of my PhD. Specifically, I will use this time to complete the specialist training outlined in the Career Development section and to build expertise in advanced data analysis and trial design. I will also conduct a literature review and audit of ITP treatment costs using Hammersmith Hospital data. These activities support Objective 3: developing an evidence-based, health economics-informed framework for microbiome-targeted therapies in ITP.

Travel and Subsistence
Travel, accommodation, and subsistence to attend key conferences in three main areas:
• Haematology – Disseminate findings to ITP clinicians and researchers, increase visibility and profile of my research and build networks, in preparation for Postdoctoral plan (future feasibility trial of microbiome-targeted ITP therapy)
• Microbiome – Present findings, stay current with research developments, and foster collaborations.
• Patient conferences – Gather feedback on study plans and ensure patient-centred dissemination of results.

Equipment
A high-performance laptop is essential for managing and analysing large-scale multi-omics datasets, including secure storage and advanced bioinformatics.

Consumables
To cover essential research and study visit-related consumables for this complex, participant-facing, multi-omics study:
Sample Collection and Processing
• Blood, urine, and stool containers, aliquot tubes, processing tubes, samples bags
• Cryovials, barcode-labelled tubes, racks, and -80°C storage boxes
• Cold chain transport: refrigerated boxes, ice packs, absorbent packaging, 95kPa bags
• PPE: gloves, lab coats, masks, aprons
Laboratory & Assay Materials
• qPCR reagents and plates, extraction kits (DNA), buffers), and purification reagents for downstream microbiome and metabolomics analysis.
• Pipette/filter tips, PCR/centrifuge tubes, plasticware
• Reagents and buffers for sample preparation and pre-sequencing clean-up
Study Visit Materials (Improving Retention & Data Quality)
• Pre-packaged self-sampling kits for stool (home delivery supported by patient feedback)
• Instructional materials, participant logs, diet/symptom diaries, feedback forms, labelling
Engagement & Communication
• Posters, lay summaries, leaflets: Professionally printed materials for recruitment, study updates, and lay dissemination.
• Printed visual aids and feedback materials for Steering Committee and patient consultation activities.
These consumables are essential for ensuring research quality, participant compliance, and a patient-centred approach. Home sampling increases the need for user-friendly, well-designed kits and communication materials.

Knowledge Mobilisation, Dissemination & Impact
Pan-London event (venue and catering costs) for study participants, ITP patients, and families. Support from two patient representatives (including travel, time, and childcare costs), to ensure inclusive and accessible communication of results.

Training and Development
Covers:
• 3 years of PhD tuition fees

• Specialist courses: R, microbiome/omics analysis, clinical trials, public involvement, EDI, and health economics
• Travel/accommodation for courses outside London (Cambridge, Exeter, Greece)
• Placement-related costs: 2 months in San Diego and 2 weeks in Toulouse

Research Inclusion
Covers translation of participant materials (PILs, posters) into four languages and interpreting services for study visits to support non-English-speaking participants.

Working with People and Communities
Covers Patient Steering Committee involvement over 3 years:
• Meetings and consultations (group and 1:1)
• Input into recruitment and results posters
• Design costs for recruitment poster
• Pan-London recruitment event: venue, catering, patient support
• Twelve online consultations with patients unable to attend in-person, to inform future feasibility trial design and acceptability

Other Direct Costs
• Reimbursement for participant travel and time, as strongly recommended by patients and Steering Committee
• Software licences: Cognitive Testing, SIMCA, GraphPad, BioRender
• Costs for specialist analysis: deep shotgun metagenomics and NMR (Novogene and National Phenome Centre)

HEI Indirect Costs
Calculated via TRAC methodology based on project FTE and Imperial College rates. Covers estates, admin, and infrastructure.

NHS Support Costs
Covers participant eligibility assessments and consent procedures by NHS staff.

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ACKNOWLEDGEMENT AND CONFLICTS

I have read and understood the terms of applying as Lead Applicant and accept this role	✓

Potential Conflicts	I have no potential conflict of interest to declare.

Lead Applicant:	Camelia Vladescu	**Application Number:**	NIHR503277
Contracting Organisation:	Imperial College London		
Proposed Start Date:	01/04/2026	**End Date:**	31/03/2029

Contributions

Narrative CV template for Personal Career Development Awards

****Lead Applicant / Co-applicant Name:** Camelia Vladescu**

****Module 1 – Contributions to the generation of new ideas, tools, methodologies or knowledge****

I began my clinical research career as a Band 5 Clinical Research Practitioner (CRP). I implemented a new master database for all research studies and an efficient patient eligibility screening system, leading to top UK and European recruitment achievements for the UK ITP Registry, TRAIT study, NIHR BioResources studies, and various clinical trials for immune thrombocytopenia (ITP).

I actively sought opportunities to lead projects. Working with Prof Cooper, I submitted my first abstract to the European Haematology Association (EHA) conference in 2018 and presented a poster on pilot data investigating cerebral microbleeds (CMBs) in ITP patients. I led patient recruitment, data collection, and contributed to analysis, review, and publication in 2020 (doi:10.1182/blood.2020004858).

Building on the CMBs study and patient reports of memory and concentration issues, I initiated a study amendment to include cognitive testing. I led patient recruitment, data collection, analysis, and writing for submission in June 2025. I presented interim and final findings at EHA 2019 (doi:10.1097/01.hs9.0000561104.19850.f7), ASH 2022 (doi:10.1182/blood-2022-166412), and the 1st European Research Consortium Meeting on ITP in Venice 2024. I was selected for the ASH Poster Walk in 2022, and the presentation was later included on a haematology teaching platform.

Engaging with over 500 ITP patients across interventional and observational studies, I noticed recurring questions about diet and gut issues. Combined with emerging studies linking the gut microbiome to autoimmune diseases, I hypothesised a potential role for the microbiome in ITP pathogenesis. Using my skills in critical thinking and patient engagement, I led the design of a pilot study on the gut microbiome in ITP. In collaboration with Professors Cooper and Marchesi, I developed the study, secured a one-year fellowship, and led design, approval, patient recruitment, and data analysis. This project allowed me to contribute new knowledge while refining my skills in protocol design, analysis, presentation, leadership, and multidisciplinary collaboration.

I have played a key role in other ITP studies through patient recruitment and data collection, contributing to two publications and six international conference presentations. Sharpened skills in study operations, data management, and scientific communication have laid the foundation for my future research independence.

I have secured competitive funding to support my research, including: Imperial Health Charity/NIHR Imperial BRC Pre-doctoral Fellowship (£83,721) to investigate the gut microbiome in patients with ITP, Imperial College London Clinical Academic Training Office Top-up Grant (£3,000), National ITP Clinical Audit Competition (£5,000) to evaluate research awareness among patients with ITP, NIHR Pre-Application Support Fund to support the development of this Doctoral Award application (£35,238).

****Module 2 – Contributions to the development of others and maintenance of effective working relationships****

Throughout my career, I have focused on developing others, fostering collaborations, and building high-performing, inclusive teams. As Team Leader for Non-Malignant

Haematology Research, I created a competency-based training programme for new staff, many without prior clinical research experience, emphasizing supervision, inclusivity, and collaboration. This contributed to the unit's strong recruitment performance and supportive culture, with several staff advancing to senior roles. I have also mentored two research practitioners pursuing clinical academic careers, offering experience and guidance on funding and career development.

As Trial Manager for the COVID-19 MATIS study, I was responsible for site setup, regulatory submissions, and multi-centre coordination. I contribute to the strategic direction of our Research Unit through its Steering Committee, guiding staff development and enhancing industry partnerships. I also represent the unit at national forums, such as the UK ITP Forum.

Recognising the importance of interdisciplinary collaboration, I established a productive relationship with Prof. Marchesi's microbiome group and the National Phenome Centre to support metabolomics for my PhD. These experiences highlight my commitment to leadership, mentoring, project management, and strategic collaborations, strengthening my ability to create supportive, high-functioning research environments.

****Module 3 – Contributions to the wider research community****

I am recognised nationally and internationally and I was invited to present at a Novartis investigator meeting in Madrid, where I shared insights on our team's patient-centred recruitment approach. This led to a second invitation to speak at a Novartis haematology education meeting in London, where I discussed the broader value of UK clinical trials, highlighting patient benefit and economic impact.

Within the UK ITP Forum, I engage in discussions on national research developments, such as the future of the UK ITP Registry. I contribute to research quality by reviewing manuscripts on ITP, including studies on CMBs and the gut microbiome.

I regularly teach healthcare professionals on clinical research. Recent examples include: (1) UK ITP Research & Update Days (May 2024) - facilitated a patient interview session; (2) Pan-London Junior Doctors Training Day (May 2024); (3) sessions for registrars, focusing on current trials, GCP training, researcher engagement and collaboration with HaemSTAR; (4) hosted a student nurse on a 6-weeks research placement and supported other nurses in accessing GCP training.

Between 2017–2020, I helped organise six international ITP preceptorships alongside Professor Cooper, including presenting on new treatments and clinical trials in 2019. In April 2025, I led a multidisciplinary session on UK regulatory approvals, employing an interactive BINGO-themed workshop to engage attendees. I consistently promote research as an inclusive, accessible, and essential part of healthcare.

****Module 4 – Contributions to broader society****

The design of this doctoral research project has been shaped by ongoing patient conversations and a diverse Patient Steering Committee, established with the Imperial Patient Experience Research Centre. This committee, reflecting various ethnic, educational, and socio-economic backgrounds, will guide the project throughout.

A key milestone in my engagement work was organising the first-ever ITP Patient Day at Hammersmith Hospital in 2017, bringing together over 50 patients and caregivers. Since then, I have co-developed and delivered annual ITP research education events

and Sickle Cell Disease (SCD) events, focusing on rebuilding trust, cultural sensitivity and improving research inclusion for this underserved community. For The SCD initiative, I secured £13,600 in funding from Genomics England and NIHR North London RDN to deliver a series of clinic-based events and one Pan-London event.

To improve public understanding of research, I contributed articles to The Platelet, the UK ITP Support Association's newsletter. My most recent piece (Nov 2024), on the potential role of the microbiome in ITP, received substantial feedback from patients (e.g. "*Thank you for committing your time to this - it matters to us ITP'ers"*).

I advocate for patient-centred research with pharmaceutical partners, ensuring fair reimbursement, especially for patients with caregiving duties or self-employment. I have also requested translations and worked to enhance patient-facing materials for inclusivity and transparency.

****Module 5 - Your suitability for this award****

My ambition is to become a clinical academic leader working at the interface of the NHS, academia, and industry, translating biologically informed, patient-centred research into real-world improvements for people with immune-mediated diseases like ITP. As one of the first nationally accredited CRPs, I am committed to mentoring others to pursue clinical academic careers and fostering a more inclusive research culture.

Since joining the NHS in 2017, I have worked solely in research delivery roles without protected academic time. Despite this, I completed an MRes while working full-time and presented at national and international meetings. In 2023, I was awarded an Imperial Health Charity/NIHR Imperial BRC Pre-Doctoral Fellowship. During this period, I set up a study on the gut microbiome in ITP, recruited patients and controls, formed a diverse Patient Steering Committee, and initiated microbiome data analysis.

The NIHR Doctoral Fellowship is a crucial step in my development. It will allow me to build on my pre-doctoral work and gain formal training in: (1) microbiome and metabolomics methods, including advanced bioinformatics and data integration; (2) health economics and intervention development to explore the cost-effectiveness of new therapies; (3) patient and public involvement and science communication to ensure my research is co-designed with, and clearly communicated to, patients.

The PhD will lay the foundation for my next career phase: applying for a **postdoctoral fellowship to design and deliver an interventional trial** on microbiome-targeted therapy in ITP. By integrating clinical outcomes, health economic data, and patient perspectives, the study will test whether microbiome modulation can improve treatment response and quality of life, and lower NHS treatment costs.

This award will provide the structured development and mentorship I need to transition from a research delivery professional to an independent clinical academic. The support of the NIHR Academy, along with guidance from my supervisory team and Patient Steering Committee, will enable me to lead collaborative, interdisciplinary research rooted in scientific rigor and patient relevance. Having navigated a non-traditional route into academia, I'm aware of the barriers CRPs and others face. I aim to use my position and the fellowship's visibility to demystify academic pathways for research delivery colleagues, creating more inclusive routes into academic leadership.

In short, the NIHR Doctoral Fellowship will prepare me to be a leader in patient-centred research that shapes the future of care in ITP and other autoimmune diseases.

Detailed Research Plan Template for Domestic Programmes and Personal Awards (maximum 10 pages)

****Background and rationale****

Immune Thrombocytopenia (ITP) is a chronic autoimmune disorder that can significantly impact daily life[1]. It is characterised by the destruction of platelets, leaving patients at risk of unpredictable bleeding, ranging from minor nosebleeds to more serious complications such as intracranial haemorrhage[2]. While it has received attention as a potential side effect of the COVID-19 vaccines[3], ITP remains a significant and growing burden on the NHS. Treatments are costly, with inconsistent responses and frequent relapses, underscoring the need for more effective, targeted, and safer therapies. Beyond physical symptoms, many patients experience chronic fatigue, cognitive impairment, and anxiety, making ITP a significant mental health challenge as well[1,4,5].

The UK incidence of adult ITP is estimated at 120 cases per year, with 3,000 - 4,000 affected at any time[6]. Numbers rose during and after the COVID-19 pandemic, and current figures are likely underestimated. For 80% of patients, ITP is lifelong.

ITP is classified as primary or secondary, the latter linked to conditions like systemic lupus erythematosus (SLE) or drug exposure (e.g. vancomycin)[7,8]. Chronic ITP often coexists with other health issues, worsening outcomes. Nordic registry data show 8.6% had a Charlson Comorbidity Index ≥3, with common comorbidities including diabetes (9.3%), solid tumours (7.9%), and hypertension (18.1%). Severe bleeding requiring hospitalisation occurred in 8.3%, and over 60% had platelet counts below 30 × 10^9/L (normal range 150-400 × 10^9/L). These findings underline ITP's role as a poor prognostic factor in multimorbid patients[9].

Primary ITP is diagnosed by excluding other causes of thrombocytopenia, leading to a diverse patient group with variable symptoms and treatment responses[10]. Although defined by platelet counts <100 × 10^9/L, treatment is usually reserved for counts ≤30 × 10^9/L or high bleeding risk, with the primary goal of reducing bleeding[8]. Bleeding in ITP is significantly more costly than in the general population due to complex treatment protocols and longer hospital stays. While NHS tariffs estimate outpatient, intracranial, and gastrointestinal bleeds at £494, £7,044, and £5,503 respectively, real-world ITP costs are much higher: £3,134, £25,699, and £14,325 respectively[11].

Beyond bleeding, ITP significantly impacts quality of life. Up to 50% of patients report fatigue and cognitive issues[4,5]. The ITP World Impact Survey found 45% reduced working hours, 29% considered quitting, and 18% retired early due to ITP[1], highlighting its psychological and socioeconomic burden.

First-line treatment for ITP is glucocorticoids; 70–80% respond initially, but only 20–30% maintain safe platelet counts after stopping. Long-term use causes side effects like weight gain, insomnia, mood changes, hypertension, and infection risk, underscoring the need for safer, longer-lasting therapies[10].

Second-line treatments for ITP, including thrombopoietin receptor agonists (TPO-RAs) (eltrombopag, romiplostim, avatrombopag) and immunosuppressants like rituximab, have variable success and often require a trial-and-error approach. TPO-RAs maintain platelet counts but lead to lasting remission in only ~20% of patients, making long-term use common. Immunosuppressants increase infection risk, need close monitoring, and often cause side effects that reduce quality of life and lead to

discontinuation. These treatments are also costly: a UK analysis estimates lifetime costs at £300,000-£400,000[11]. While they reduce bleeding, they don't target the underlying autoimmunity, leaving symptoms like fatigue unresolved. Their cost, side effects, and limited durability highlight the need for safer, more sustainable treatment options.

Emerging role of the gut microbiome in ITP

What is the gut microbiome?

The gut microbiome is the community of microbes and their genes found in the digestive tract. It plays a crucial role in maintaining immune balance and overall health. This system includes both the microbes themselves (the microbiota) and the collection of metabolites (chemicals) they produce (the metabolome). Both the microbes and their metabolites, such as short-chain fatty acids and bile acids, can influence immune function. The composition of the gut microbiome is shaped by multiple factors, including diet, environment, medication use (e.g. antibiotics), and lifestyle[12].

Studies of the gut microbiome typically measure the proportion of each microbe relative to the total community (known as relative abundance), rather than the actual number of microbes present. To estimate total microbial quantity, a method called quantitative PCR (qPCR) can be used. Microbial diversity is often summarised using alpha diversity indices (which reflect how many different types of microbes are present in a single sample) and beta diversity indices (which compare differences in microbial composition between samples)[13]. In addition to analysing the microbes themselves, metabolomic profiling examines the microbial-derived metabolites, helping to understand how changes in the microbiome may affect host biological processes[14].

Given the microbiome's role in maintaining immune equilibrium, it is perhaps unsurprising that disturbances in its composition and function (known as dysbiosis) have been implicated in the pathogenesis of several autoimmune diseases, suggesting that it may influence immune responses by either promoting tolerance or contributing to autoimmunity [15–18]. This growing understanding has led to increased interest in microbiome-targeted interventions, including probiotics (live beneficial microbes), prebiotics (compounds that promote the growth of beneficial microbes), dietary changes, and faecal microbiota transplantation (FMT), which aim to restore or modulate microbial balance[12].

What is the role of the microbiome in ITP?

Although the microbiome has not been extensively studied in ITP, preliminary evidence suggests potential links. For example, *Helicobacter pylori* infection (bacteria which alters the gut microbiome) has been associated with increased platelet counts following eradication, particularly in patients from Japan and Italy. Responses are less common in severe or long-standing cases[19]. Anecdotal reports also suggest that antibiotics, probiotics, and dietary changes (e.g., intermittent fasting) may influence platelet levels.

Several small observational studies have reported gut microbiome differences between ITP patients and healthy controls, such as increased *Pseudomonadota* and altered *Bacteroidota* to *Bacillota* ratios [20–23]. However, findings are inconsistent, with some studies reporting the opposite or no significant differences, including mixed results on alpha diversity [21,24,25]. These discrepancies may reflect small sample sizes, differences in microbiome analysis methods, and limited use of absolute bacterial quantification via qPCR. Most studies were conducted in China, limiting

generalisability to UK populations. Few included well-matched controls, which is crucial given the influence of diet and environment on the microbiome. Including "ecologically" related controls, such as household members, helps distinguish ITP-specific differences from those driven by shared external factors.

While early evidence suggests a microbiome-ITP link, inconsistent findings highlight the need for robust, well-powered studies in diverse populations (including the UK) using standardised protocols to clarify the microbiome's role in ITP.

Pre-doctoral research

As part of my pre-doctoral fellowship (2023–2024), I initiated a pilot study comparing the gut microbiome of 18 ITP patients and 4 related ("ecological") healthy controls, selected to minimise dietary and environmental confounding. I supplemented controls with data from an external cohort using the same 16S rRNA gene sequencing. Analysis included microbiota profiling (16S rRNA gene sequencing and qPCR) and metabolite detection via proton nuclear magnetic resonance (NMR) spectroscopy in samples with complete data. A demographically diverse group of patients and controls was included (Table 1).

Table 1. Demographics and clinical characteristics of the population studied.

	ITP patients (N=18)	**Related healthy controls (N=4)**	**All healthy controls (N= 39)**
	Median (range)	Median (range)	
Age, years	43 (19,65)	33.5 (24.5, 44)	34 (21, 56)
Female, n (%)	13 (72)	2 (50)	22 (56)
Ethnicity			
Asian or Asian British	4 (22)	2 (50)	18 (46)
Black, Black British, Caribbean or African	1 (5.5)	0	0
Mixed or multiple ethnic groups	3 (17)	0	0
Other ethnic group	0	0	2 (5)
White	10 (55.5)	2 (50)	19 (49)
Duration of ITP, months	126 (4,572)	N/A	N/A
Number of prior ITP therapies	3 (1,18)	N/A	N/A
Nadir platelet count (lowest count since diagnosis)**,** x 10^9/L	2 (0,13)	N/A	N/A

Findings (microbiome diversity):

- ITP patients had reduced microbiome diversity (Figure 1).
- Alpha diversity was significantly lower in patients with a platelet nadir <5×10^9/L (p=0.04) and in those in those diagnosed with ITP for more than 12 months (p=0.04)
- Alpha diversity was significantly lower in patients with cognitive impairment (p=0.05)
- Beta diversity differed significantly between ITP patients and the larger cohort of healthy controls (p=0.001), indicating distinct microbial community composition.

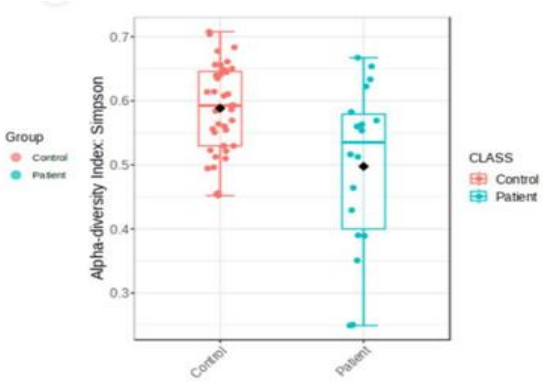


Figure 1: Alpha diversity was significantly lower in ITP patients (n=18) compared to healthy controls (n=39) (p=0.01).

Consultations with patients: During a patient engagement event participants (N=13) were asked which interventions they would be willing to try if research identified a link between the gut microbiome and ITP. Twelve participant expressed willingness to try prebiotics/probiotics, while (7/13) 54% were open to undergoing FMT. Responses highlight strong patient support for continued microbiome research and openness to novel therapeutic approaches (Figure 2).

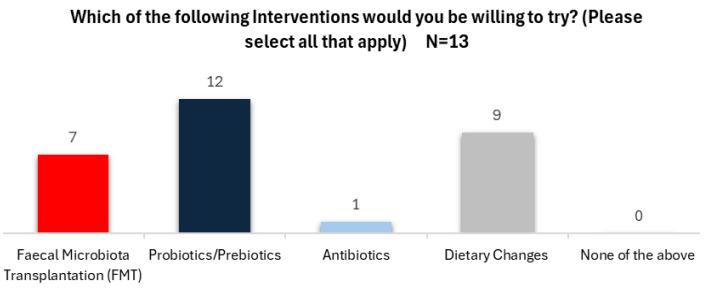


Figure 2: Patient Willingness to Try Microbiome-Modulating Therapies. Results from a patient engagement event exploring interest in potential gut microbiome interventions for ITP.

Implications for treatment: Microbiome dysbiosis may be present in ITP and contribute to differences in patient outcomes. Restoring microbial balance through interventions like pre/probiotics or Faecal Microbiota transplant (FMT) could reduce reliance on the costly or poorly tolerated current treatments, improving outcomes, quality of life, and healthcare costs.

Addressing the aims of the NIHR Doctoral Award: This project addresses an unmet clinical need in ITP through early translational research, exploring the gut microbiome as a novel, non-immunosuppressive therapeutic target to improve outcomes and reduce treatment burden.

****Aims and objectives****

Research aim: My aim is to establish the contribution of gut microbiome dysbiosis in ITP in order to target it for a cost saving therapeutic intervention.

Key objectives:

Objective 1: Compare gut microbiome profiles between adults with ITP and related healthy controls.

Objective 2: Explore associations between microbiome features and ITP severity, symptoms, and response to treatment.

Objective 3: Evaluate whether insights from gut microbiome profiling can inform future therapeutic strategies for patients with ITP.

****Methodology / Plan****

Objective 1: How does the gut microbiome differ between adults with ITP and healthy controls?

Preliminary data from my pre-doctoral fellowship showed that patients with ITP have reduced gut microbiome diversity. During this PhD, I will build on these findings by further characterising the gut microbiome in ITP and comparing it to that of age- and sex-matched healthy family controls.

Study design: I will conduct a cross-sectional, case-control study comparing the microbiota and metabolic profiles of patients with ITP and related “ecological” controls.

Study population: Adult patients with primary ITP (platelet count ≤30 x 10^9/L at any time during their disease duration) and their household relatives.

Sample size: Based on predoctoral fellowship data (n=18 ITP patients, n=39 healthy controls), alpha diversity (Simpson index) was lower in patients (mean = 0.498, SD = 0.129) than in controls (mean = 0.590, SD = 0.070) (p=0.01), with an estimated effect size of 0.87 (Cohen's d). A two-sample t-test indicated that 23 participants per group would provide 80% power at a 5% significance level. A threshold-based approach, using the lowest alpha diversity among controls (0.45) as a cut-off, found 33% of patients fell below this threshold (6/18), compared to 0/39 controls. Power analysis using this binary outcome also supported a sample size of 23 per group. However, due to reliance on stool samples for DNA extraction and sequencing, issues like insufficient sample volume or technical failures may arise. To account for potential challenges, based on experience form the pilot study, the sample size will be increased by 30%, and 30 ITP patients and 30 age- and sex- matched controls will be recruited.

Study procedures

- One hospital visit per participant to collect stool, blood and urine samples (stool to assess gut microbiota composition; all samples for metabolite analysis)
- Clinical assessments: bleeding questionnaire (ITP patients only), CANTAB cognitive test, quality of life and fatigue questionnaires, 7-day food diary
- Medical history review and extraction from patients' electronic medical records

Inclusion & Exclusion Criteria

Inclusion: (1) Primary ITP patients: platelet ≤30x10^9/L at any disease stage; (2) Healthy volunteers: household members of the patients enrolled in the study.

Exclusion: Antibiotic use in the past month; pregnancy or breastfeeding; history of moderate/severe gastrointestinal (GI), hepatic or cardiovascular disease, diabetes, metabolic syndrome, bowel surgery, IBD, IBS, or other GI disorders; autoimmune disease (controls only); current diagnosis of malignancies; age under 18*.

*In the pilot study, we set an upper age limit of 65, common practice in many microbiome studies, based on the assumption that older participants would likely have comorbidities and take multiple medications (polypharmacy). Following input from the Patient Steering Committee, this limit was removed, as many over-65s are healthy and age alone was not seen as a justified exclusion. Comorbidity-based criteria remain.

Objective 2: Are specific microbiome profiles associated with disease severity, clinical symptoms or treatment response in ITP?

During my pre-doctoral fellowship, I identified that patients with more severe disease (platelet count <5 x10^9/L) and longer ITP duration had lower alpha diversity. Reduced microbiome diversity was also linked to impaired cognition.

I will further assess links between the microbiome and disease severity, symptoms, and treatment response by grouping patients by platelet count, number of therapies, bleeding, ITP duration, and features like fatigue and cognitive impairment. I will also repeat study procedures in 10 patients from my pre-doctoral cohort to explore longitudinal changes in microbiome and clinical features and potential causal links.

Data collection and analysis (Objectives 1&2)

The primary research question is whether gut microbiome diversity differs between ITP patients and healthy controls. The primary outcome is alpha diversity (including Shannon, Simpson and Chao 1 indices), calculated from shotgun metagenomic sequencing of stool samples. Statistical comparisons will primarily use non-parametric

tests, such as the Mann–Whitney U test, due to the typically non-normal distribution of microbiome data. Multivariate regression will adjust for potential confounders such as age, sex, diet, and medication use. Although controls are matched on age and sex, these variables will still be included as covariates where relevant.

Secondary outcomes include beta diversity (Bray-Curtis dissimilarity, visualised via PCoA), taxonomic and functional profiles (relative and absolute abundances obtained by qPCR), and metabolite patterns from NMR spectroscopy. PCA will assess overall variance; PLS-DA will identify group-discriminating metabolites.

Subgroup analyses will compare diversity and composition across ITP groups (e.g. platelet count, disease duration, treatment response, fatigue, cognition, bleeding). Tests will include ANOVA/Kruskal-Wallis, chi-square/Fisher's exact, and Pearson/Spearman correlations.

Exploratory longitudinal analysis in 10 patients from the pilot study will assess changes in microbiome and clinical features over time. Given the small sample, these analyses will be descriptive and hypothesis-generating.

Sensitivity analyses will explore the impact of key confounders (e.g. comorbidities). All analyses will be performed using R and SIMCA.

Objective 3: Can gut microbiome insights inform future therapeutic strategies for ITP?

This project will lay the groundwork for a future interventional trial by developing a patient-informed, evidence-based framework for microbiome-targeted therapies in ITP. While observational, the study will explore both therapeutic potential and patient acceptability.

In Year 1, I will conduct a structured literature review and audit NHS costs (via Hammersmith Hospital data) to assess the cost-effectiveness of current ITP treatments. I will build on existing collaborations with the Centre for Health Economics and Policy Innovation and attend relevant training to incorporate economic evaluation into future trial design.

In parallel, I will expand on pre-doctoral work to gather patient views on microbiome-targeted therapies by:

- Hosting a patient event in Year 1 to support recruitment and begin education and early consultation with patients on microbiome-based therapies
- Presenting findings (from Objectives 1 and 2) to the Patient Steering Committee for feedback on future trial design and intervention preferences/acceptability
- Organising a dedicated patient engagement event in Year 3 involving study participants, their families, carers, and other individuals affected by ITP. The event will be hybrid so patients from across the country will be invited to attend via the UK ITP Support Association. The event will include two components: (i) dissemination of Objective 1 &2 findings, and (ii) a facilitated group discussion to explore acceptable microbiome-based interventions and practical considerations for future trials
- Attending external events such as the annual UK ITP Patient Association Convention
- Conducting up to 12 one-on-one virtual consultations for wider input

Findings from microbiome profiling, patient engagement, and preliminary cost-effectiveness insights (via literature review and local NHS audit) will be synthesised to

guide future feasibility trial design, including intervention selection, key clinical and patient-reported outcome measures, and participation barriers. Early input from health economists will ensure the trial is built with cost-effectiveness in mind.

Justification for choice of methodology

Objectives 1 &2: This is a cross-sectional, case-control study powered to characterise microbiome abnormalities in a UK ITP population, generating clinically relevant data to guide the development of targeted microbiome-based interventions.

I chose to include ecological controls (household members) who share similar dietary and environmental exposures to control for confounders, increasing the likelihood that observed microbiome differences are disease-related. This approach is supported by my Patient Steering Committee.

For high-resolution microbial profiling, I will use deep shotgun metagenomic sequencing, the gold standard in microbiome research[26]. Untargeted NMR metabolomics, chosen for its reproducibility and cross-study comparability, will complement sequencing[27]. Quantitative PCR (qPCR) will be used to enable absolute bacterial quantification, addressing the limitation of relative abundance in metagenomic data. Together, these methods offer a comprehensive view of microbiome composition, function, and host interactions[14].

Using validated tools (SMOG bleeding score[28], CANTAB cognitive testing[29], FACIT fatigue[30] and ITP-PAQ quality of life questionnaires[31]), I will link microbiome profiles to modifiable symptoms and treatment response, informing patient selection, outcomes, and trial design for future microbiome-based ITP interventions.

Objective 3: A structured literature review and the case study on treatment costs at Hammersmith Hospital will clarify the economic context of ITP treatments. Patient engagement through events, one-to-one interviews and Patient Steering Committee discussions, will ensure diverse views are captured. These methods are appropriate to develop a patient-informed, evidence-based translational framework grounded in clinical and real-world priorities.

Timelines and milestones

I obtained regulatory approvals for my study during my pre-doctoral fellowship, so recruitment can start immediately.

Year	Activities	Milestones
1	- Participant recruitment, study visits, and sample collection - Invite 10 patients who participated in the pilot study for a follow-up visit - Deliver first Pan-London event to boost recruitment and start education and consultation on microbiome-based therapies - Conduct structured literature review on health economics of ITP treatments and start to audit ITP treatment costs at Hammersmith Hospital - Begin training in microbiome methods, R/stats, leadership, EDI, and PPIE - Early-Stage Assessment	- Recruitment of 30 pairs - Follow-up visits for Objective 2 completed -Complete literature review - Learn how to analyse the data (metagenomics and metabolomics)
2	- Complete sample processing and data cleaning - Conduct detailed bioinformatic and statistical analyses for Objective 1 & 2 - Continue patient and public involvement: • Present results to Patient Steering Committee • Attend patient engagement events and webinars - Advanced training in statistical modelling and science communication - Prepare abstracts for conferences and manuscripts for publication - Late-Stage Assessment	- NMR and shotgun metagenomics data ready for analysis (Month 14) - Final results for Objectives 1 & 2 analysed (Month 24) - Conference abstract submissions (Month 24)

3	- Conduct qualitative data collection and analysis for Objective 3: • Engagement event and one-to-one interviews with patients • Synthesis of patient preferences on microbiome-targeted therapies - Integrate microbiome and health economic insights to design a translational framework for microbiome interventions - Finalise all analyses and interpret findings - PhD thesis courses (Imperial) - Prepare PhD thesis and submit manuscript for peer-reviewed journals - Disseminate findings at patient and scientific conferences - Participate in grant writing and career development training	- Qualitative engagement completed including one to one interviews and patient engagement event (Month 30) - Translational framework developed (Month 32) - Thesis draft completed (Month 34) - Thesis submission and viva (Month 36)

Working with people and communities & Inclusive research

This research question was shaped by my conversations with hundreds of ITP patients (from various demographic and socioeconomic backgrounds) during recruitment and study visits for various research studies. Many shared gut-related experiences, such as stomach infections before diagnosis, or asked whether diet could influence their condition. Some reported that fermented foods or intermittent fasting seemed to raise their platelet counts. Before applying for my pre-doctoral fellowship, I also consulted six patients from diverse backgrounds to provide feedback on the initial study idea and to shape the project.

During the fellowship, I worked with Maria Pigging from Imperial Patient Experience Research Centre (PERC) to set up a Patient Steering Committee. It includes six people with ITP (one being Dianne White, Deputy CEO of the UK ITP Patient Support Association) and two family members. With PERC's support, I designed an application process to ensure a diverse group was selected. The Committee has been involved from the start, shaping both the pre-doctoral and PhD research. Their feedback influenced key elements of this proposal, including the strategies to recruit related healthy controls, reduce hospital visits by using stool home collection kits, and remove an upper age limit to ensure inclusivity. They also contributed to and reviewed the Plain English Summary.

My goal is to reduce barriers to taking part in research. I have already hosted patient-friendly events (e.g. at Hammersmith Hospital), where patients showed strong interest in microbiome-based therapies. I will continue engaging diverse groups through the UK ITP Patient Support Association. I am committed to equity and inclusion throughout this PhD, following best practices in inclusive research[32].

Key activities include:

- Study materials (including the Patient Information Leaflet and a one-page poster to advertise the study) will be reviewed for plain language and cultural relevance by the Patient Committee and translated (if required). Translators will be asked to interpret during research visits if required. One patient required a translator in the pilot study.
- In the first six months, I will host a large patient engagement and recruitment event at Chelsea Football Club (a familiar and accessible venue that can attract a diverse audience across demographic and socio-economic backgrounds). Alongside myself, two patient collaborators will help present the research in an accessible way and answer questions. To ensure a welcoming atmosphere and accommodate attendees' needs, the event will be fully catered and designed with comfort and accessibility in mind.
- In Years 2 and 3, I will present emerging findings to the Patient Steering Committee and co-create a patient-facing summary of the results.

- In Year 3, I will host a second patient event to share the results of the study with those who participated in the project, as well as any other patients or families affected by ITP. I will also consult them on the acceptability of microbiome-based interventions and practical considerations for the future interventional trial.
- I will share the results and gather feedback through ITP Support Association events and up to 12 one-to-one patient consultations to shape the future interventional study.
- Research results will be shared in different ways (online, in print, and at events) to reach people with different levels of health knowledge and digital access.

I will also continue training in inclusive and ethical research to ensure this work is shaped with, and for, people living with ITP.

****Knowledge mobilisation, dissemination and impact****

The intended outcomes of the research are to: (1) Establish whether patients with ITP have microbiome dysbiosis; (2) Determine whether microbiome-targeted therapies are acceptable to patients with ITP; (3) Inform the design of a feasibility study for a microbiome-based intervention in ITP.

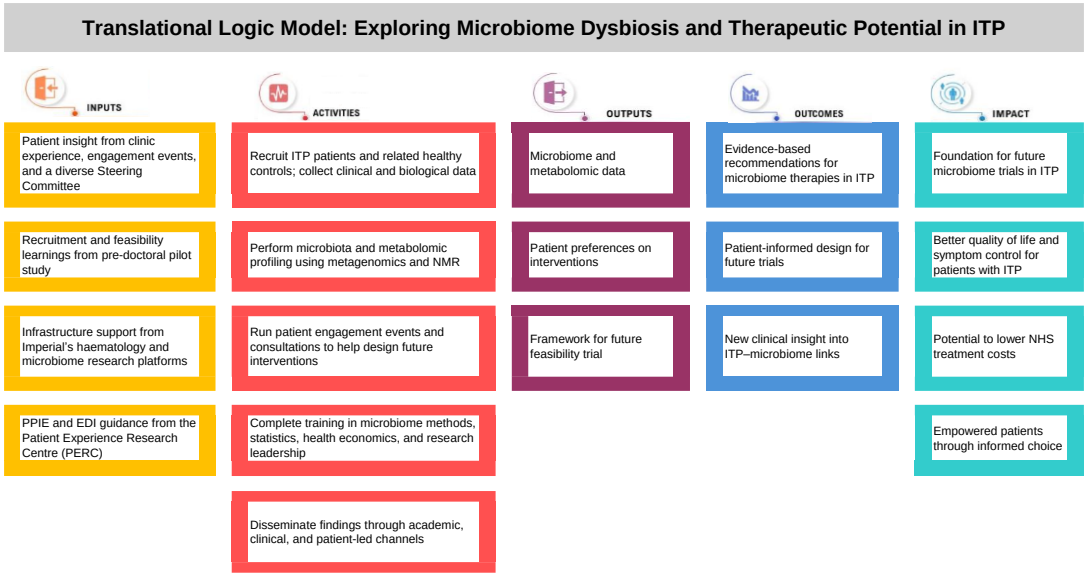
The anticipated impact of this work is: (1) **For patients**: Address a long-standing question about the gut's role in ITP, and support development of more personalised, less immunosuppressive treatments to improve symptoms and quality of life; (2) **For the NHS**: Identify microbiome-based strategies that could reduce treatment duration, reliance on high-cost drugs, hospital visits, and overall costs of treating ITP.

Primary knowledge users include people with ITP, their families, and the clinical teams treating them. Other stakeholders include microbiome and immunology researchers, biotech/pharmaceutical companies developing microbiome-based interventions, policy makers and advocacy groups (e.g., the UK ITP Patient Support Association).

Knowledge mobilisation is embedded across all stages of the research:

- **Study Design**: I developed the project with input from patients (via clinics, events, and the Patient Steering Committee), supervisors, microbiome/immunology research groups (via our regular meetings), haematologists and Clinical Nurse Specialists (via NW London MDT, ITP Forum, and ITP Update Days), the UK ITP Patient Support Association, NIHR Research Support Service, and the Non-malignant Haematology Research Unit at Hammersmith Hospital.
- **Ongoing Engagement:** I will maintain regular engagement with all stakeholders involved in the study design by presenting interim and final analyses to all key stakeholders (at the same meetings/events included above).
- **Dissemination:** Results will be shared at national and international haematology and translational science meetings (British Society of Haematology, American Society of Haematology, The European Haematology Association, European research consortium on ITP), and published in peer-reviewed journals. The study will be registered and results reported on the ISRCTN Registry. Findings will be shared with patients and the wider community through the ITP Support Association's communication channels (including social media, printed newsletters, and in-person events), a dedicated participant dissemination event in Year 3, and a co-produced one-page summary (distributed both in ITP clinics and online).
- **Translational Impact:** I will pursue engagement with pharmaceutical/ biotech companies to explore opportunities for future interventional studies.

I will collect both qualitative and quantitative indicators of reach and impact: (1) Feedback and attendance at patient engagement events; (2) Reach of digital materials (e.g. social media posts metrics); (3) Number of conference abstracts, invited talks, and publications; 4) Testimonies from patients, clinicians, and industry partners about the value of findings.



****Project management****

Recruitment and Study visits: Recruitment will be led in collaboration with the Immune Haematology Clinic at Hammersmith Hospital, supported by the Non-Malignant Haematology research team. I will screen patients, with the research team coordinating contact, stool kit delivery, and visits. To minimise burden, participants will receive study information by phone and attend a single clinic visit, with stool kits posted in advance. Recruitment pathways: (1) Hammersmith Hospital clinic; (2) National promotion via the UK ITP Patient Association (e.g. events, The Platelet magazine); (3) Referrals from other centres (PIL and posters will be shared and study discussed at MDTs and national meetings); (4) A Pan-London recruitment event in Year 1.

Clinical and Research Integration: I will attend weekly meetings with the Immunology and Microbiome Research Groups at Imperial to stay up to date with the latest scientific developments, share project updates and receive feedback. I will attend regular Non-malignant Haematology research team meetings to coordinate recruitment and study visits.

Supervision and Oversight: I will meet with my supervisors every 1-2 weeks to review progress against Gantt Chart, troubleshoot challenges, and adapt plans as needed.

Patient Steering Committee: I will hold group meetings at least twice a year and maintain contact via phone/email every other month, as outlined in the Gantt Chart.

****Intellectual property and commercialisation****

This PhD may generate Intellectual Property (IP), including microbiome/metabolomic signatures linked to ITP and a patient-informed framework for intervention design. Imperial College London will retain IP ownership. No patents have been filed, and no third-party IP or external licences are needed. All tools are open-source or institutionally licensed. Future partnerships with biotech or pharma may be pursued if results support intervention studies.

Career Development Programme for Personal Career Development Awards (maximum 3 pages)

****Training courses****

Guided by the Vitae Researcher Development Framework, this training and development plan supports my growth across key domains of research competence and professional practice. It includes a mix of short and in-depth courses delivered by Imperial College London and external providers, offering diverse perspectives and broader networking opportunities. See Gantt chart for course timing.

1. R Programming and Statistical Analysis: Learning R and relevant statistical techniques is essential for conducting metagenomics and metabolomics data analysis. I will complete online courses: *Data Science: R Basics* (Harvard University) and *Introduction to the Microbiome R Package*. I will attend in-person courses at ICL (*Foundations in R* and *Introduction to Statistics Using R)* to complement my coding skills with statistical applications.

2. Advanced Microbiome Data Analysis: To analyse metagenomic and metabolomic data, I will take advanced courses including *Introduction to Metagenomics Analysis* and *Metagenomics Bioinformatics at MGnify* at EMBL-European Bioinformatics Institute, and the *Metagenomics Data Analysis Course* at the University of Cambridge. I will complete online courses like *SIMCA for Metabolomics* (Sartorius) and the *Novogene Webinar Hub* and attend the *Gut Microbiome Summer School* hosted by the University of Ioannina and ICL.

3. Clinical Trials and Research Delivery: To strengthen my CRP skills and prepare for designing future microbiome trials, I will attend the *ICTU Contemporary Clinical Trials Course* at ICL, focusing on advanced trial design and statistics Additionally, I will complete the *Clinician Researcher Credentials Framework Postgraduate certificate* via Kings College London (NIHR).

4. Health Economics: To develop skills for postdoctoral grant applications involving cost-effectiveness, I will attend: *Economic Evaluation: an Introduction* and *Understanding Health Economics in Clinical Trials* at University College London.

5. Scientific Writing and Literature Review: I will take the *Literature Review* course at ICL, which will aid in writing a review on health economics in ITP. To support my manuscript and thesis writing, I will enrol in the *Writing in the Sciences* course (Stanford University online) and attend ICL's *Planning & Preparing for Your Thesis & Viva* and *Thesis Writing Retreat*.

6. Working with People and Communities: To improve my communication and engagement skills, I will attend the *PPIE in Health and Care Research Summer School* at the University of Exeter/NIHR and complete the *Public Involvement Training* at ICL.

7. Equality, Diversity, and Inclusion (EDI): To ensure inclusive research design and leadership, I will attend the following courses: *Explore Equality, Diversity and Inclusion* (CIPD), *Inclusive Research* (Integrated Interdisciplinary Innovations in Healthcare Science Hub), *Learning Lectures: Making Research More Inclusive* (NIHR), *Unconscious Bias* and *The Power of Diversity and Inclusion in Research* (ICL).

8. Science Communication: To improve my ability to communicate research to diverse audiences and understand its impact on society, policy, and the economy, I will attend *Research Impact Workshops* at ICL.

9. Leadership and Research Management: To support my transition to postdoctoral independence, I will enrol in the *ARC South London Applied Research Leadership Academy* and in *Leadership Webinars and E-learning Modules* (NIHR). Online courses will include: *Researcher Management and Leadership Training; Strategic Leadership: Impact, Change and Decision-Making* (Coursera), *Enhance Your Leadership Skills* (ICL).

This training program will equip me with the technical, methodological, leadership, and collaborative skills to complete my doctoral project and pursue a postdoctoral career in translational microbiome research for immune disease.

****Placements****

To strengthen my international networks and my expertise in microbiome research, clinical trial design, and translational research, I plan to undertake three placements.

1. Prof Rob Knight's Interdisciplinary Metagenomics Group (University of California, San Diego) – 2 months

Prof. Knight is a global leader in microbiome science and director of the UC San Diego Centre for Microbiome Innovation. This placement will deepen my understanding of cutting-edge metagenomics methodologies, computational microbiome analysis, and their translational applications in human health. During the placement, I will have my own data ready for analysis and will directly apply the methods and insights gained to my research. Working in Prof. Knight's group will allow me to receive targeted feedback on my analysis and results, and me help build international collaborations.

2. Prof Guillaume Moulis (Toulouse University Hospital, France) – 2 weeks

Prof. Moulis, an international expert in ITP and director of the pharmacoepidemiology unit at Toulouse University Hospital, coordinates the referral centre for adult autoimmune cytopenias in Toulouse. During this placement, I will gain insight into the design and analysis of large-scale studies and real-world data in ITP. I will also explore opportunities for international research collaboration and data harmonisation in rare autoimmune conditions.

3. Industry Engagement (Paddington Life Sciences, London) – 2 weeks

Paddington Life Sciences brings together NHS, academic, and industry partners to accelerate research translation. I will complete a placement under the mentorship of the Director of Strategic Partnerships to gain experience in engaging with commercial partners, and navigating NHS-industry collaborations, enhancing my knowledge of health innovation and strategic partnerships for leading translational research in ITP.

****Building Collaborations****

During the PhD, I will continue developing existing interdisciplinary national and international collaborations in microbiome science, immunology, data science, patient involvement, and translational research. This will ensure the project is scientifically rigorous, patient-centred, and positioned for real-world impact, supporting both the PhD and my goal of leading interventional research in ITP.

As an already embedded member of Prof Marchesi's group, I will continue to work with Dr. Chrysostomou (postdoc) who will oversee the microbiome data analysis throughout my PhD. Dr Amna Malik (postdoc) in Prof Cooper's immunology group will provide bioinformatic support to analyse large datasets and Prof will offer guidance on clinical trial design and modelling treatment costs in ITP. Throughout my PhD, I will

share my progress and data with both research groups to gather feedback, validate my analyses, and ensure the study stays on track.

To broaden the impact of my research, I will develop new collaborations. I will engage with the European Research Consortium on ITP (ERCI) to build international partnerships for future trials, disseminate findings to ITP clinicians, and influence future ITP guidelines.

Attending the ICTU Contemporary Clinical Trials Course will build relationships with statisticians, trial design experts and health economists, with the aim of co-developing the design of a future interventional trial after my PhD. I will participate in the launch of the Imperial Health Economic Evaluation Initiative to further strengthen these relationships.

Working with people and communities will continue to be a cornerstone of my work. I will collaborate closely with the Patient Experience Research Centre (PERC) at Imperial to stay up to date with best practices in inclusive and meaningful PPIE and EDI. I will also involve the Immunology BRC patient group in co-designing the proposed microbiome-targeting trial, supporting my postdoctoral application.

****Conferences****

Attending national and international conferences will provide valuable opportunities to disseminate results, receive feedback from experts, stay updated on the latest developments in ITP and microbiome research, and build academic and clinical collaborations. I will attend major conferences like the *European Haematology Association (EHA), British Society for Haematology (BSH), and American Society of Haematology (ASH)* to disseminate my findings and engage with global experts and potential collaborators. Additionally, I will participate in *Imperial College London's Microbiome Network Annual Conference* to share early data and gain feedback from colleagues specializing in gut health and host-microbe interactions.

I will also attend the *European Research Consortium on ITP (ERCI)* and *ITP Update Day (UK)* to present my findings to a highly relevant clinical and academic audience. To stay informed on analytical methods and developments in microbiome research, I will engage with the *Human Microbiome Conference at the Wellcome Genome Campus* and the *London Microbiome Meeting (TwinsUK)*. Furthermore, the *British Society of Gastroenterology's Gut Microbiota Conference* will help me keep up to date with the latest research on the gut-immune interface and emerging microbiome-targeting therapies.

Finally, I will continue to be an active participant in the *Clinical Academic Training Office (CATO) Annual Meetings and Symposia* to enhance my development in research leadership and communication while connecting with other Clinical Research Practitioners (CRPs) pursuing academic pathways.

****Other****

I will continue collaborating closely with the Non-Malignant Haematology Research Unit and stay in regular contact with my line manager, attending research and haematology meetings with the team. This will help me stay informed about new developments and the evolving requirements of the Clinical Research Practitioner (CRP) profession. Additionally, I will co-deliver sessions on research delivery and career development for early career CRPs through the Imperial CATO, enhancing visibility and supporting others pursuing clinical academic careers.

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Head of Department
Director of Bioservices
Professor & Honorary Consultant in Rheumatology

06 June 2025

Dear Members of the Selection Committee,

RE: NIHR Doctoral Training Fellowship – Camelia Vladescu

I am writing to offer my full support for Camelia Vladescu's application for the NIHR Doctoral Fellowship.

Camelia joined Imperial College London in 2023 when she secured competitive funding from the Imperial Health Charity and the Imperial Biomedical Research Centre (BRC) to undertake her pre-doctoral fellowship. Within the Department of Immunology and Inflammation, Camelia has been a valued member of our research community, showing exceptional initiative, intellectual curiosity, and a commitment to advancing the understanding of immune thrombocytopenia (ITP). Her exciting research outputs include a publication in *Blood* (2022) on cognitive impairment in ITP patients and a number of poster presentations, including at the American Society of Haematology (ASH) Annual Meeting and the British Society of Haematology Annual Meeting. Camelia's active participation in the academic community goes beyond presenting her work. She has also been proactive in securing research funding to support her work, with notable grants including the Imperial Health Charity/NIHR Imperial BRC Pre-doctoral Fellowship (£83,721), a £5,000 award from the National ITP Clinical Audit Competition sponsored by Grifols and a £32,731 grant from the NIHR Pre-application Support Fund as part of the NIHR Career Development Scheme for Doctoral Fellowships. These accomplishments reflect her ability to secure competitive funding, a crucial skill for an independent researcher.

Camelia's proposal builds on her previous research into the microbiome abnormalities in patients with ITP, by focusing on a highly novel and under-explored area: the role of microbiome dysbiosis in ITP. Her work will be supervised by Professor Nichola Cooper, an internationally recognized expert in ITP (in both basic science and translational research), and Professor Julian Marchesi, a leading authority in microbiome research. Her research bridges two major NIHR Imperial BRC themes: immunology (co-led by Prof Cooper) and digestive diseases (co-led by Prof Marchesi). This interdisciplinary approach, combining immunology with microbiome research, positions Camelia's work to have a transformative impact on the field of autoimmune disease treatment.

Imperial College London is fully committed to supporting Camelia's ongoing professional development. The university offers a wide range of professional development resources, including access to its award-winning Early Career Researchers Institute, which encompasses the Graduate School, and the Clinical Academic Training Office which provides tailored support for clinical academics. Camelia will also benefit from dedicated mentorship, a variety of training opportunities, and access to world-class research facilities. As part of the Imperial BRC, Camelia will continue to benefit from the guidance of senior academics and clinicians and be provided with opportunities to expand her existing network. Camelia is already embedded in Professor Cooper's research group and has contributed meaningfully to its research activities. During her fellowship, she also established a strong and collaborative relationship with Professor Marchesi's microbiome group, which has directly informed the development of her doctoral proposal. Camelia's ability to build productive, cross-disciplinary relationships and work seamlessly across clinical and academic teams is a clear strength, and will be instrumental in the success of her PhD and in her development as a future leader in translational, patient-centred research.

We are confident that Camelia will emerge from this fellowship as an independent researcher capable of making significant contributions to clinical research in ITP and related autoimmune disorders. Beyond the fellowship, there are clear opportunities for her continued development at Imperial, including potential post-doctoral roles supported by the NIHR Imperial Biomedical Research Centre (BRC) and the Clinical Academic Training Office (CATO), both of which are committed to nurturing post-doctoral researchers across all professional groups.

Additionally, Camelia will enhance her public engagement skills by developing and leading workshops in collaboration with patient organisations such as the ITP Support Association, which has already funded part of her pre-doctoral project. These workshops will allow her to engage directly with patients, refine her skills in clinical trial design, and strengthen her relationships with pharmaceutical companies and other stakeholders in the ITP field. This will provide her with a comprehensive skillset that spans both research and public engagement, preparing her for leadership roles in clinical research.

At Imperial we are deeply committed to fostering an inclusive and supportive research environment. Camelia's project will benefit from the collaborative and diverse research culture within the Department of Immunology and Inflammation, as well as the Department of Digestion, Metabolism, and Reproduction. Both departments have a proven track record of nurturing junior researchers into independent academic leaders. Our commitment to equality, diversity, and inclusion (EDI) is reflected in our Athena SWAN awards, participation in the Race Equality Charter, and our status as a Disability Confident Employer. Camelia's research will be conducted within this inclusive environment, where she will have the opportunity to work alongside researchers from various professional and cultural backgrounds, ensuring that her research is enriched by diverse perspectives. The university also adheres to the highest standards of research integrity. We have rigorous procedures in place for ensuring ethical conduct, data management, and compliance with regulatory guidelines. Camelia will receive training in research ethics and integrity as part of her professional development and will continue to operate within these standards throughout her fellowship.

In conclusion, I confirm my full and enthusiastic support for Camelia's NIHR Doctoral Fellowship application. Camelia has shown strong academic ability, a proactive approach to research, and a commitment to public engagement that will allow her to thrive as a future leader in the field of autoimmune disease research. We are confident that her work will have a significant impact on the treatment of ITP and will contribute to the growing body of knowledge on the role of the microbiome in health and disease.

Yours sincerely,

Marina Botto

Professor Marina Botto (Head of Department of Immunology and Inflammation)

Imperial College Healthcare
NHS Trust

09/06/2025

Dr Abdul Shlebak MBChB (Hon) MD FRCP FRCPath DIML
Consultant Haematologist and Clinical director for Clinical Haematology at
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Partner Organisation Letter of Support – NIHR Doctoral Fellowship Application for Camelia Vladescu

As Head of the Department of Haematology at Imperial College Healthcare NHS Trust (ICHNT), I am delighted to offer our full support for Camelia Vladescu's NIHR Doctoral Award application. Camelia, a Band 7 Clinical Trial Practitioner, is the Team Leader of our Non-Malignant Haematology Research Unit. She is a very bright, hard-working team leader who has led this team from strength to strength and has demonstrated exceptional leadership and research capabilities. She is an exceptional early-career academic with a clear trajectory toward independence, which makes her an outstanding candidate for this award.

Under Camelia's leadership, our Non-Malignant Haematology Research Unit has grown to become one of the largest in the UK. The team frequently ranks as top recruiters for pharmaceutical-sponsored clinical trials in the country and has received commendations for their outstanding performance, including achieving the first patient recruited in Europe and the UK. Camelia's impressive leadership has yielded results that have earned her invitations to speak at national and international events, where she shares her strategies for recruiting patients to clinical trials and building a research culture. Her team's contributions have been instrumental in the licensing of four ground-breaking drugs for haematological conditions, including immune thrombocytopenia (ITP).

In 2023, Camelia was awarded a competitive pre-doctoral fellowship from the Imperial Health Charity and NIHR Imperial BRC, which she completed in September 2024. Her research on microbiome abnormalities in ITP was highly successful and formed the basis for her proposed PhD project. She showed considerable initiative in developing this novel research area across specialties, culminating in a poster presentation at the annual Imperial BRC conference. In 2024, we supported Camelia's application for the NIHR Pre-Application Support Fund, which she was awarded and commenced in December 2024. This award enabled her to develop this Doctoral Fellowship application while continuing to lead the Research Unit three days per week.

During this time, Camelia led a successful bid for Genomics England and NIHR North London RRDN funding to deliver research education and engagement events for underserved communities affected by sickle cell disease. Working within a tight timeframe, she built a multidisciplinary team of healthcare professionals, patients, and advocates to organise the first clinic-based engagement events at our hospital, as well as a Pan-London event. These were well received by patients and colleagues, with outstanding feedback gathered through surveys and shared on social media channels by participants and collaborators. The success of this initiative led to further funding to expand the programme. This work reflects Camelia's strong leadership, her ability to coordinate complex, inclusive research activities, and her commitment to patient-centred engagement, which are hallmarks of an emerging clinical academic leader.

Her proposed doctoral fellowship on the role of the gut microbiome in immune thrombocytopenia (ITP) builds directly on the promising preliminary data generated during her pre-doctoral fellowship. ITP is a complex condition that can be challenging and costly to manage, and Camelia's research addresses a significant unmet clinical need. By taking a translational, cross-disciplinary approach, her project has the potential to inform future therapies that are not only more effective and better tolerated but also more cost-efficient. This aligns strongly with ICHNT's strategic priorities to support innovative research that improves patient outcomes and reduces healthcare burden.

Support During the Fellowship

ICHNT is fully committed to supporting Camelia's fellowship. She will have access to Trust facilities, clinical data systems, laboratories, and experienced research staff across the Trust and Imperial College London (ICL). She will be supported by the Non-Malignant Haematology Research Unit, the haematology clinical team at Hammersmith Hospital, as well as her line manager and our Lead Nurse for the Clinical Research Workforce.

Camelia will continue to benefit from regular engagement with her supervisors, Professor Nichola Cooper and Professor Julian Marchesi, through the NIHR Imperial BRC's Immunology and Digestive Diseases Themes, which span both ICHNT and ICL. Professor Cooper, who is the lead of the Immune Haematology service at ICHNT (the largest service of its kind in the UK), is an academic consultant in the Centre of Haematology within the Department of Immunology and Inflammation. A world-renowned expert in ITP, she treats the largest ITP patient cohort in the UK and co-chairs the European Research Consortium on ITP. She also plays a central role in designing and running clinical trials in this area, which will provide Camelia with invaluable experience in trial design and implementation, particularly in the context of ITP. This experience will be pivotal for her planned future work in designing and leading interventional trials as part of her postdoctoral research. Professor Marchesi, a leading authority in Digestive Health and the co-founder of the Microbiome Network, was the chair of the British Society of Gastroenterology's Gut Microbiota for Health Expert Panel from 2018-2021 and chaired the Microbiology Society's "Unlocking the Microbiome" working group and the subsequent publication.

Camelia's project spans two NIHR Imperial BRC research themes: Immunology (theme co-led by Prof Cooper) and Digestive Diseases (theme co-led by Prof Marchesi). This positions her work at the forefront of a novel and clinically relevant area that has the potential to reshape future treatment strategies for ITP. Both departments have strong records of supporting early-career researchers toward academic independence and will provide a rich, multidisciplinary environment for Camelia's continued development.

Post-Fellowship Support

Beyond the fellowship, ICHNT is committed to supporting Camelia's transition to an independent clinical academic career. Through the Clinical Academic Training Office (CATO), she will continue to access training, mentoring, and structured career planning support, including the Healthcare Professionals Academic Group (HPAG), grant writing workshops, and ongoing mentorship from our Director of Clinical Research Workforce. She will also receive bespoke career advice from Professor Mary Wells, our Director of Nursing Research.

Camelia has already expressed a clear intention to apply for a postdoctoral fellowship following completion of her PhD. We will support her in identifying and preparing for appropriate postdoctoral funding opportunities, ensuring she is well-positioned to build on her doctoral research. ICHNT has previously supported several NMAHPs to transition from NIHR Fellowships into postdoctoral and clinical academic leadership roles, and we are confident Camelia will successfully follow this trajectory.

Inclusive Research Culture & EDI

We are strongly committed to an inclusive, supportive research culture, with equity, diversity and inclusion (EDI) embedded across our work with staff, patients and communities. Our 2024 anti-racism strategy and use of equality impact assessments reflect a systemic, data-driven approach to promoting fairness and representation. We support career progression through initiatives like the Healthcare Leaders Fellowship, diverse recruitment panels, and staff networks, and track progress via diversity dashboards. As a Level 2 Disability Confident Employer and member of the Employers Network for Equality and Inclusion (ENEI), Stonewall, and the Business Disability Forum, the Trust fosters an accessible, inclusive workplace, offering guaranteed interviews for disabled applicants and mental health training for managers.

Camelia's project, which includes robust public and patient involvement and specifically addresses diversity in recruitment and engagement, aligns fully with these institutional priorities. Our Trust encourages inclusive research design, and Camelia's approach to co-developing study materials with patients, and remove unnecessary age restrictions embodies these principles. She will continue to champion EDI throughout her fellowship by participating in staff networks, promoting inclusive research practices, and contributing to PPIE (Patient and Public Involvement and Engagement) initiatives.

Research Integrity

ICHNT upholds the highest standards of research integrity, with oversight from our Research Governance Office and alignment with national frameworks. Camelia is well-versed in GCP and ethical conduct and will be supported by our teams to ensure all research adheres to regulatory and ethical standards.

Summary

Our vision at ICHNT is to be a world leader in clinical academic careers for nurses and allied healthcare professionals, ensuring that patient care is underpinned by evidence. Camelia is an outstanding candidate who has already demonstrated initiative, leadership, and innovation. Her proposed doctoral project is timely, translational, and patient-focused, with clear potential to address unmet clinical needs in non-malignant haematology through microbiome research.

We are fully committed to supporting Camelia throughout this fellowship and beyond. She will be embedded in a dynamic research environment with access to multidisciplinary expertise, structured academic career development, and strong mentorship. Her passion for inclusive, evidence-based research aligns closely with the Trust's priorities, and we are confident in her trajectory toward becoming a national leader in this field. We are proud and delighted to support her in achieving her future career goals.

Abdul Shlebak

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8th June 2025

To Whom It May Concern,

I am writing to offer my strongest support for Camelia Vladescu's application for the NIHR Doctoral Fellowship. I have had the pleasure of working closely with Camelia since 2017 when she joined our team as a junior Clinical Research Practitioner. Camelia is an exceptionally hard-working, resourceful, and driven clinical academic who consistently strives to exceed expectations.

It has been immensely rewarding to see her rapid progression from a junior Clinical Research Practitioner to leading our Non-Malignant Haematology Research Unit. She is an exceptionally hard-working leader who has guided this team to new heights, showcasing remarkable leadership and strong research acumen that make her an ideal candidate for this award.

Under Camelia's guidance, our Non-Malignant Haematology Research Unit has grown into one of the largest units in the UK. The team consistently ranks among the top recruiters for pharmaceutical-sponsored clinical trials in the country and has received numerous commendations for their exceptional performance, including multiple national and European firsts in trial recruitment.

Camelia's outstanding leadership has led to notable achievements, earning her invitations to speak at national and international conferences where she shares her expertise in patient recruitment for clinical trials and fostering a research-oriented culture. Her team's contributions have been pivotal in the licensing of four new treatments for non-malignant haematology conditions, including immune thrombocytopenia (ITP).

Despite having no protected time for academic development before 2023, she successfully completed an MRes while working full-time during the COVID-19 pandemic and has contributed to multiple peer-reviewed publications and international conference presentations. For the project she initiated during her MRes, she presented both interim and final results at major haematology conferences, including American Society of Haematology 2022, British Society of Haematology 2024, and the European Research Consortium on ITP (ERCI) in 2024. This work has culminated in the submission of a full manuscript in June 2025, further demonstrating her ability to deliver high-quality research outputs and engage effectively with the academic community.

In 2023, Camelia was awarded the NIHR Imperial BRC/Imperial Health Charity Pre-Doctoral Fellowship, which she successfully completed under my supervision. During the 12 months of her fellowship, she set up a pilot study investigating the gut microbiome in patients with ITP. She also collaborated with the Imperial Patient Experience Research Centre to recruit a diverse Patient Steering Committee for her study, which has been actively involved throughout her pre-doctoral fellowship and in shaping her Doctoral Award application. Her ability to collaborate

effectively across departments, set up the project, including writing the protocol, patient information materials, and all other required documents, recruit participants, perform the study visits, and successfully analyse the data (despite having no prior experience in microbiome data analysis) further underscores her exceptional dedication, meticulous approach, intellectual curiosity, and readiness for doctoral-level research.

Camelia's proposed doctoral research builds directly on this strong clinical foundation and her prior pre-doctoral fellowship. Her project, which I will supervise, addresses an urgent unmet clinical need: current treatments for ITP are limited to thrombopoietin receptor agonists (TPO-RAs) and immunosuppressive therapies, both of which present significant challenges. TPO-RAs are expensive and typically require lifelong use, while immunosuppressants leave patients vulnerable to infections and other complications. Despite these options, a substantial proportion of patients do not respond adequately or experience only partial, short-lived benefit—some remain refractory to all available treatments. Moreover, these therapies do not address the burdensome symptoms of fatigue and cognitive impairment that persist even when platelet counts are stabilised. Camelia's project investigates whether gut microbial changes may influence immune dysregulation, disease burden, and treatment response in ITP. It combines deep shotgun metagenomics, untargeted metabolomics, validated symptom assessments, and meaningful public and patient involvement to design a patient-informed, evidence-based framework for future microbiome-targeted interventions in ITP. The proposal is innovative, feasible, and aimed at improving patient outcomes, quality of life, and reducing NHS treatment costs for the ITP patient population.

I am particularly impressed by how Camelia has embedded patient voices throughout the research, from shaping inclusion criteria and study materials, to planning knowledge mobilisation and acceptability studies of future microbiome-targeted interventions. This is an exemplary translational programme: it is grounded in real patient concerns, informed by cutting-edge science, and designed to lay the groundwork for a future feasibility trial with health economic integration.

Our supervisory relationship is already well-established and long-standing. Camelia and I have been meeting regularly, often more than once a week, since she joined Imperial in 2017. Throughout her doctoral training, this close working relationship will continue. We will continue holding dedicated supervision meetings every week, with monthly joint sessions that include her co-supervisor, Professor Julian Marchesi, which started during her pre-doctoral fellowship. In addition, Camelia will continue attending weekly pre-clinic meetings and research team meetings within the Non-Malignant Haematology group, as well as weekly lab meetings with my research group. She will also maintain her close collaboration with the Non-Malignant Haematology Clinical Research Unit, who will support with participant recruitment and study visits, and with the Immune Haematology clinical team at Hammersmith Hospital. I will continue to supervise her doctoral work closely, contributing my expertise in immune haematology, clinical trial methodology, and translational research, while providing structured and day-to-day mentorship throughout the PhD.

Camelia's PhD is directly aligned with my groups research focus and will be based within my groups highly active research environment and nationally recognised immune haematology service, the largest ITP service in the UK.

I am a Professor of Immune Haematology at Imperial College London and co-lead the Immunology Theme for the NIHR Imperial BRC. As a founder of the European Research Consortium on ITP, and trustee of the UK ITP patient support association, I lead national and international initiatives in this field. My research group focuses on the immunopathogenesis of

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haematological autoimmunity, and I run one of the UK's largest ITP clinical cohorts and trial portfolios.

I have successfully supervised three PhD students to completion as well as numerous MSc and BSc students within haematology and immunology, all of whom have achieved excellent academic results and presented their work at national and international meetings. I currently supervise two PhD students in basic science projects as well as two clinical research fellows and two postdoctoral fellows, and I provide clinical leadership for the clinical trials team.

My group runs one of the most extensive ITP trial portfolios nationally, as national and international CI. I personally lead and design phase 1–3 studies in this area and have developed strong collaborations with the Imperial Clinical Trial Unit, and the Imperial Business School. Camelia will be embedded within this highly active clinical-academic environment. She will also benefit from co-supervision by Professor Julian Marchesi, a leader in microbiome research, and co-lead of the Digestive Diseases Theme for the NIHR Imperial BRC. Therefore, she will and access to world-class infrastructure through the Imperial BRC's Immunology and Digestive Diseases Themes, both of which align directly with her project.

In conclusion, Camelia is a highly capable, innovative, and committed early-career researcher who has demonstrated all the qualities of a future leader in clinical academic research. Her strong academic background, proven research track record, and dedication to patient-centred science make her an outstanding candidate. Her proposed project is well aligned with NIHR's translational and patient-focused priorities, and I am confident she will deliver this PhD with distinction and make a meaningful contribution to the field. I wholeheartedly support her application for the NIHR Doctoral Fellowship.

Yours sincerely,

Sincerely,

Professor Nichola Cooper MA, MD, FRCP, FRCPath
Professor of Immune Haematology
Director of the Centre for Immune Thrombocytopenia
Divisional Director of Research Surgery and Cancer
Co-lead, Immunology NIHR BRC theme
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Professor Julian R Marchesi PhD
Professor of Digestive Health

5th June 2025

Re: Support for Ms Vladescu's application

To whom it may concern:

I am writing in strong support of Camelia Vladescu's application for the NIHR Doctoral Award. I have supervised Camelia throughout her pre-doctoral fellowship, funded by the Imperial Health Charity and NIHR Imperial Biomedical Research Centre (BRC), and during the preparation of this Doctoral Fellowship application, supported by her NIHR Pre-Application Support Fund. I am fully committed to continuing as her co-supervisor throughout her doctoral studies and beyond.

Camelia is a committed and well-organised early-career researcher who has demonstrated the potential to establish herself as an independent clinical academic investigator. She has taken initiative in shaping her doctoral proposal and shown strong leadership, particularly in cross-sector collaborations and inclusive research engagement. Her project aligns closely with NIHR's translational priorities and will benefit from our multidisciplinary team, including Professor Nichola Cooper's world-leading expertise in ITP.

We have already established an effective and collaborative supervisory relationship, meeting monthly during her pre-doctoral fellowship and maintaining open communication between formal meetings. Camelia has also become a valued participant in my research group's lab meetings, where she has developed strong connections with other team members and gained practical exposure to microbiome data analysis and research techniques. This has laid a solid foundation for her continued learning and integration into our team.

Camelia's proposed project (examining the role of the gut microbiome in immune thrombocytopenia (ITP)) is highly relevant to my expertise. Her study takes a translational, cross-disciplinary approach to an underexplored clinical condition and has the potential to inform future microbiome-targeted interventions. The project sits at the intersection of two NIHR Imperial BRC themes: Digestive Diseases, which I co-lead, and Immunology, co-led by Professor Nichola Cooper. This unique alignment will allow Camelia to benefit from the infrastructure, training opportunities, and networks of both BRC themes, further strengthening the translational potential of her research.

A particular strength of Camelia's work is her robust commitment to Patient and Public Involvement and Engagement. She independently established a diverse Patient Steering Committee comprising individuals with ITP from a range of demographic and socioeconomic backgrounds. I had the opportunity to attend some of these committee meetings. We discussed both the findings from her pre-doctoral work and potential therapeutic avenues, including microbiome-targeting strategies. The conversations were frank and constructive, and patients openly engaged with the idea of microbiome modulation and were receptive to exploring the feasibility of interventions such as faecal microbiota transplant (FMT) in ITP. This level of patient engagement reflects Camelia's ability to facilitate inclusive, transparent, and forward-thinking research dialogue.

I am a Professor of Digestive Health in the Faculty of Medicine at Imperial College London, with over 24 years of research experience in microbiome science. My work integrates metagenomics, metatranscriptomics, metabonomics, and molecular ecology to understand host–microbiome interactions. I have authored over 280 peer-reviewed publications (H-index 78 (SCOPUS)) and held senior editorial roles, including at the International Society for Microbial Ecology Journal. I have led national working groups, including the British Society of Gastroenterology's Gut Microbiota for Health Expert Panel (2018–2021) and the Microbiology Society's "Unlocking the Microbiome" initiative.

I have supervised 42 PhD students, all of whom completed within the allotted 4-year timeframe. I currently supervise 14 PhD students, alongside two postdoctoral fellows and two research technicians. I also have significant experience mentoring MSc and BSc students. I have a strong track record in supporting early-career researchers through to independence, which I will bring to Camelia's supervision. I will continue to meet monthly with Camelia throughout the fellowship, with additional ad hoc support as needed. As her co-supervisor, I will contribute to her training and development plan, guide her through microbiome methodologies, and support her in preparing publications, conference presentations, and future funding applications.

Beyond this doctoral fellowship, I am committed to supporting Camelia's transition into post-doctoral research and long-term academic development. Our research aspirations are strongly aligned: my group is currently involved in clinical trials of microbiome-targeting therapies, and we are in the process of establishing an NIHR-funded FMT production facility. Camelia's project builds directly into these efforts, and I look forward to mentoring her as she progresses toward applying for post-doctoral fellowships and contributing to future intervention studies in this space.

Your Sincerely,

Professor Julian R. Marchesi

UNIVERSITY *of* CALIFORNIA, SAN DIEGO
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June 5, 2025

To Whom It May Concern,

I am pleased to support Camelia Vladescu's proposed short-term placement in my laboratory at the University of California, San Diego, as part of her NIHR Doctoral Fellowship.

Camelia is undertaking a translational research project investigating the role of the gut microbiome in immune thrombocytopenia (ITP). Her project integrates multi-omics approaches, including deep shotgun metagenomic sequencing and untargeted metabolomics, to investigate microbiome-host interactions in a rare autoimmune condition. I understand her pre-doctoral work has already generated promising pilot data and engaged a diverse patient population through an inclusive research approach.

During her proposed two-month placement in my lab, Camelia will receive practical training in advanced metagenomic data analysis, learning to apply high-throughput workflows developed by the Knight Lab. She will be supported by our multidisciplinary team, which specialises in the computational and experimental techniques necessary to characterize microbiomes across diverse ecosystems, including the human body. Her training will focus on areas such as metagenomic data processing, statistical interpretation, and integrating multi-omics datasets, with an emphasis on reproducible and open-access bioinformatics methods.

Our group has long been at the forefront of microbiome science. I am Professor of Pediatrics, Bioengineering, Data Science, and Computer Science & Engineering at UC San Diego and founding Director of the Center for Microbiome Innovation. My lab developed many of the foundational tools that have shaped the field, including the QIIME pipeline and UniFrac, and co-founded large-scale efforts such as the Earth Microbiome Project and the American Gut Project. Our research has contributed to understanding microbial involvement in health and disease, including obesity and inflammatory bowel disease, and made microbiome analysis accessible to thousands of scientists globally. I am also a Fellow of the American Association for the Advancement of Science and the American Academy of Microbiology and was awarded the Vilcek Prize in Creative Promise for the Life Sciences in 2015 and elected to the National Academy of Engineering in 2024.

Camelia's clinical insight, commitment to inclusive research, and growing technical capability make her an excellent candidate for this placement. We are confident she will benefit from and contribute to our collaborative environment. Her time with us will strengthen the analytical foundation of her PhD and foster connections that may support future international research partnerships.

I fully support her visit and look forward to welcoming her to UC San Diego, should her NIHR Doctoral Fellowship application be successful.

Sincerely,

Professor Rob Knight, Ph.D.
Wolfe Family Endowed Chair in Microbiome Research at Rady Children's
Director, Center for Microbiome Innovation
Professor, Department of Pediatrics, Department of Computer Science & Engineering, Shu Chien-Gene Lay Department of Engineering, and Halıcıoğlu Data Science Institute, UC San Diego
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Object: support letter

To: whom it may concern

I, Professor Guillaume Moulis, MD, PhD, coordinator of the Toulouse referral center for autoimmune cytopenias in adults and director of research at the pharmacoepidemiology unit of the Clinical Investigation Center of Toulouse University Hospital, France, fully support the Doctoral research program of Camelia Vladescu, under the supervision of Professor Nichola Cooper (Imperial College London).

We would be happy to welcome her to our institution for a short research placement as part of her PhD. As an international expert in immune thrombocytopenia (ITP) and director of the pharmacoepidemiology unit, I am delighted to host this placement, which will provide Camelia with hands-on exposure to the design and analysis of large-scale studies and real-world data in ITP. During the placement, Camelia will have the opportunity to explore avenues for international research collaboration and data harmonisation in ITP. These experiences will directly support the aims of her doctoral research and contribute to her development as a clinical academic in the field of autoimmune diseases.

I look forward to welcoming Camelia to Toulouse and supporting her training and research development.

Prof. Guillaume Moulis

Project Timeline

YEAR 1 (April 2026- March 2027)

		2026									2027		
		April	May	June	July	August	September	October	November	December	January	February	March
Objective 1 & 2 activities	Participant Recruitment & Sample Collection												
	Sample Processing & Shipping												
	Medical history (data) collection												
Objective 3 activities	Literature review (health economics of ITP)												
	Cost of treating ITP at Hammersmith Hospital audit												
Patient engagement	Organise patient engagement events												
	Recruitment and engagement event (Objectives 1, 2 &3)						Pan London Education and recruitment event						
	Patient Steering Committee Meetings (Objectives 1, 2 &3)												
	Patient Steering Committee check-in via phone/email												
Training	R Programming and Statistical Analysis												
	Advanced Microbiome and Omics Data Analysis												
	Clinical Trials and Research Delivery												
	Health Economics												
	Scientific Writing and Literature review												
	Working with People and Communities												
	Equality, Diversity and Inclusion (EDI)												
	Science communications courses												
	Leadership and Research Management												
	PhD Early-stage assessment												
	Placement+							Paddington Life Sciences Placement (London)					
Building collaborations & networking	Conferences*		UK ITP Update day 2026	Micorbiome Annual Conference & CATO Research Symposium 2026					ERCI 2026		London Microbiome Meeting 2027		Microbiome Interactions in Health and Disease & British Society of Gastroenterology's Gut Microbiota Conference 2027
	Patient Conventions		ITP Support Association Annual Patient Convention 2026					Annual ITP Patient Day at Hammersmith Hospital					
Supervision and mentorship	Supervisor meetings												
	Weekly research meetings (immunology and microbiome groups)												
	Mentor meetings												

YEAR 2 (April 2027 - March 2028)

		2027									2028		
		April	May	June	July	August	September	October	November	December	January	February	March
Objective 1 & 2 activities	Sample analysis (Deep shotgun metagenomics & NMR)												
	Data Cleaning & Pre-processing												
	Data Analysis - Objective 1 & 2												
Objective 3 activities	Cost of treating ITP at Hammersmith Hospital audit												
Patient engagement	Patient Steering Committee Meetings												
	Patient Steering Committee check-in via phone/email												
	Organise patient engagement event												
Training	Advanced Microbiome and Omics Data Analysis												
	Science communications courses												
	Leadership and Research Management												
	PhD Late-stage assessment												
	Placement+				Prof Rob Knight's group (San Diego)								
Building collaborations & networking	Conferences*			EHA & Microbiome Annual Network Conference 2027					ERCI 2027		London Microbiome Meeting 2028		Microbiome Interactions in Health and Disease & British Society of Gastroenterology's Gut Microbiota Conference 2028
	Patient Conventions		ITP Support Association Annual Patient Convention 2026					Annual ITP Patient Day at Hammersmith Hospital					
Supervision and mentorship	Supervisor meetings												
	Weekly research meetings (immunology and microbiome groups)												
	Mentor meetings												

YEAR 3 (April 2028 - March 2029)

		2028									2029		
		April	May	June	July	August	September	October	November	December	January	February	March
Objective 3 activities	Data Synthesis & Interpretation (after patient engagement events)												
	Design of Future Feasibility Study												
Patient engagement	Organise patient engagement events												
	Dissemination and engagement event (Disseminate Objectives 1& 2 results and consult on Objectives 3)	Pan London Dissemination and engagement event											
	One to one consultations with patients (Objective 3)												
	Patient Steering Committee Meetings (Objectives 1, 2 &3)												
	Patient Steering Committee check-in via phone/email												
Training	Scientific Writing (incl Thesis Writing Retreat)												
	Science communications courses												
	Leadership and Research Management												
	Placement+					Prof Guillaume Moulis (Toulouse)							
Building collaborations & networking	Conferences*	BSH 2028	UK ITP Update day 2028	EHA & Microbiome Network Annual Conference 2028					ERCI 2028	ASH 2028	London Microbiome Meeting 2029		Microbiome Interactions in Health and Disease & British Society of Gastroenterology's Gut Microbiota Conference2029
	Patient Conventions		ITP Support Association Annual Patient Convention 2028					Annual ITP Patient Day at Hammersmith Hospital					
Dissemination of results	Manuscript Preparation & Submission												
	Conferences*	BSH 2028	UK ITP Update day 2028	EHA & Microbiome Network Annual Conference 2028					ERCI 2028	ASH 2028	London Microbiome Meeting 2029		
	Patient Conventions		ITP Support Association Annual Patient Convention 2028										
	Dissertation write-up										Thesis draft completed		
	Post doc fellowship application												
Supervision and mentorship	Supervisor meetings												
	Weekly research meetings (immunology and microbiome groups)												
	Mentor meetings												

*Any additional conference costs not covered by the award will be sought from other sources (e.g. CATO bursary, pharma partners)

+During my placements, I will continue progressing on other aspects of my PhD as outlined in the Gantt Chart. Specifically, during the 2-month San Diego placement, I will have my data collected and ready for analysis. I will begin data analysis during this placement, applying the techniques and skills I am acquiring throughout the placement.

During my pre-doctoral fellowship, I obtained regulatory approvals for the study so recruitment can start immediately. Before starting the PhD, I will co-design a recruitment poster with the Patient Steering Committee, using remaining pre-doctoral funding, to enhance recruitment and patient engagement. I will also apply for an ethics amendment to include the poster.

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